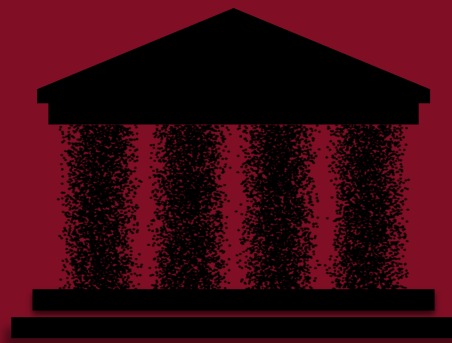

Handbook of
Special Relativity



Phy-Hub

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Intro

Classical physics accurately predicts the movement of objects in our everyday world. However, there are effects it does not account for, and as objects approach the speed of light, these effects, which are negligible at low speeds, become increasingly significant. This leads to two observers moving at high speeds relative to each other becoming unable to agree on the lengths of objects and the times at which events occur. Crucially though, one thing remains constant: the speed of light relative to each observer is the same, regardless of their movement. We must redefine our classical understanding of time and space to find out how this can be. This is where special relativity comes into play.

Special relativity is vital in particle physics, nuclear physics, and astrophysics, used to calculate nuclear reaction output power and measure galaxy speeds. It is also needed for precision timekeeping on fast-moving GPS satellites to accurately calculate positions.

Despite its importance, special relativity continues to confuse even those well-versed in physics and mathematics. It can be as confusing to visualize and understand as quantum mechanics is. The theory requires letting go of some deeply held intuitions about the absolute nature of space and time, which differ from our everyday experience of the physical world.

This handbook is meant to offer an intuitive, visual, and detailed overview of special relativity. The aim is to stay simple and avoid unnecessary abstraction in explanations. For example, the diagrams in this handbook represent time

in one of two ways: through animations or through a storyboard of snapshot diagrams that display a timeline of events. This approach avoids the use of unnecessary space-time diagrams that represent time on one of the axes, which make special relativity more abstract and confusing to visualize.

The first chapter is readable even for those without a math or physics background. It provides a non-mathematical, visual overview of the concepts. This chapter builds an intuitive understanding of the core ideas step by step, providing a solid foundation for developing the essential mathematical framework in later sections. I drew inspiration from many resources, which includes the book "Relativity visualized" [4].

The mathematical portion will begin with deriving how space and time coordinates function in special relativity. This will be followed by velocity addition, the Doppler effect, and the energy-momentum relationship. In addition to these topics, we will also explore some interesting subjects not typically covered in standard resources. These include how a source's emitted light is concentrated in the direction of its movement, aberration, and how a delay in light signals to an observer means they are currently viewing sources of light in their past positions. These topics will probably not mean much to you yet, but they will give fascinating insights.

Now, let us begin!

Part I

The Concepts Explained

Here, we will give an overview of the main concepts of special relativity, without the math. This is to show how special relativity works without needing to comprehend the math at the same time, as the math can distract from understanding the concepts themselves.

Chapter 1

Overview of Concepts

To make sense of special relativity, we will first look at the classical understanding of the world. Then, we will see how we must change these classical laws of physics to account for light's constant speed relative to all observers. Which in turn will lead to many interesting consequences. The chapter begins with an overview of the core concepts to help us understand why we need special relativity and what it means, leaving the mathematical details and deeper insights for the following chapters.

This chapter's concepts will establish the mathematical framework for later chapters by using physical objects, such as moving trucks on roads, to represent the perspectives of different observers. Whereas in the mathematical chapters, we will use coordinate axes to represent the different perspectives (frames of reference).

Later, you will see the parallels between this conceptual overview and the mathematical chapters. This chapter will provide an overview of all the required concepts for special relativity, building up the needed knowledge bit by bit, starting with our classical understanding of how perspectives (frames of reference) work and showing how our understanding must change if we are to allow all observers to view light as moving at the same constant speed.

1.1 Classical Addition of Velocities

Imagine a truck carrying a cannon moving at 20 meters per second, and the cannon fires a ball at 100 meters per second relative to the truck. Classically, we add the truck's speed to the ball's speed, to find that the ball moves at 120 meters per second relative to the road. So, in classical physics, velocities are directly added. The ball relative to the road would move at the speed of the ball relative to the truck plus the truck's speed relative to the road, as shown in Figure (1.1).

However, this classical understanding is just a very good approximation for objects in our everyday life that we are used to seeing, which are moving much slower than the speed of light (roughly 300,000,000 meters per second), but we will get to this later. For now, in the following sections, we will explain what an inertial reference frame is and how this works in the classical context so that you can see more clearly the differences between classical and special relativity.

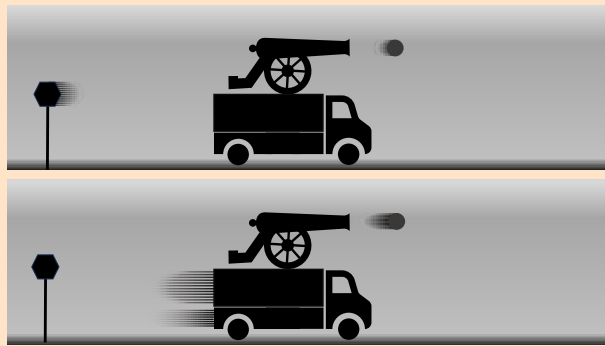


Figure 1.1: **Classical addition of velocities.** A diagram, showing the speed of a cannonball from two different perspectives: (top) from the truck's perspective, with the cannon at rest and the road moving backwards, the cannonball is shot and moves forward. The second figure (bottom) is from the road's perspective, with the truck moving relative to it. When the cannonball is shot, it looks like it is moving forward faster, with the previous cannonball's speed combined with the speed of the cannon from which it was shot.

1.2 Inertial Reference Frame

A reference frame is a grid of coordinates we can overlay on space, so that we can state, with respect to it, where and when an event occurs. If this grid is stationary or moving at a constant velocity, it is called an inertial reference frame. In the previous truck example, we could associate a reference frame with the truck by introducing a grid of coordinates that is stationary relative to the truck and, hence, moving along with it. This perspective, fixed to the truck, is the truck's frame of reference. A second grid of coordinates, stationary relative to the road, can also be introduced to represent the road's reference frame. These grids of coordinates are shown from the road's reference frame in Figure (1.2).

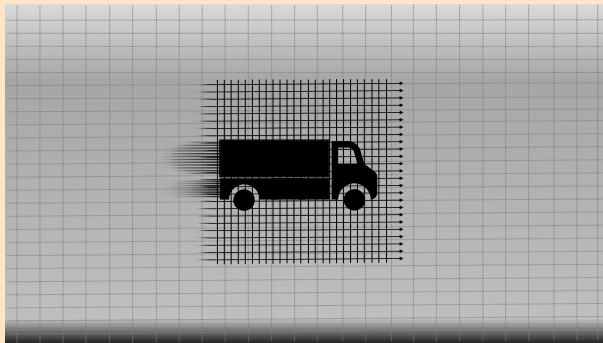


Figure 1.2: **Reference frames of a truck and road.** A diagram, showing the reference frame of a truck as a grid of coordinates that is moving with the truck relative to the road's reference frame, shown as a stationary dashed gray grid of coordinates.

An inertial frame of reference is a reference frame that is not being accelerated. You can tell if you are being accelerated, as you will feel a force. For example, in an accelerating car, you will feel the seat accelerate into you, with your body pushing back into it. This principle is used in accelerometers to measure acceleration, as shown in Figure (1.3).

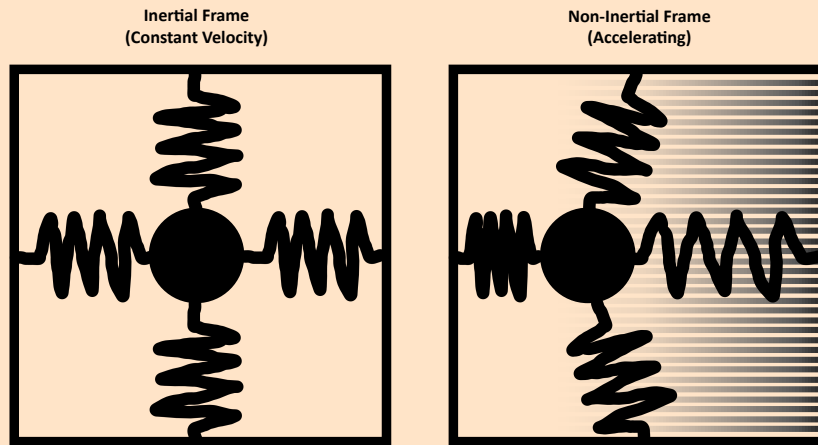


Figure 1.3: **Inertial versus non-inertial frames.** A diagram, showing a ball attached to the walls of a box by springs, with the ball centered in the box in the inertial frame, with no acceleration (left), and in a non-inertial frame (right) where the box is now accelerating to the right. The ball lags behind as the box accelerates.

We will now look at how movement changes in classical relativity when we change between these inertial frames.

1.3 Classical Inertial Reference Frames

To see how to switch between reference frames in special relativity, we first need to introduce the classical approach.

To illustrate this, we will consider a setup with two rats on a treadmill, as shown in Figure (1.4), with three rocks hanging above the platform, which are at rest relative to the room. The two frames of reference will be the treadmill's platform and the room in which it is located. Both rats start under the same rock. One rat runs to a rock positioned in the treadmill's forward direction. The other rat runs to an equally spaced rock to the side. After this, they return to the starting rock.

Here, the platform can be seen as the medium in which the rats move. Both rats travel at the same constant speed relative to the platform, with the

platform either at rest or moving more slowly than the rats, so they can reach the rocks.

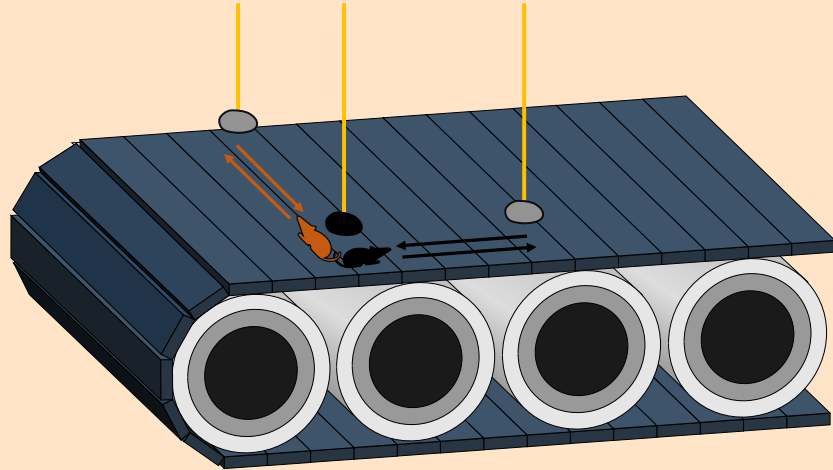
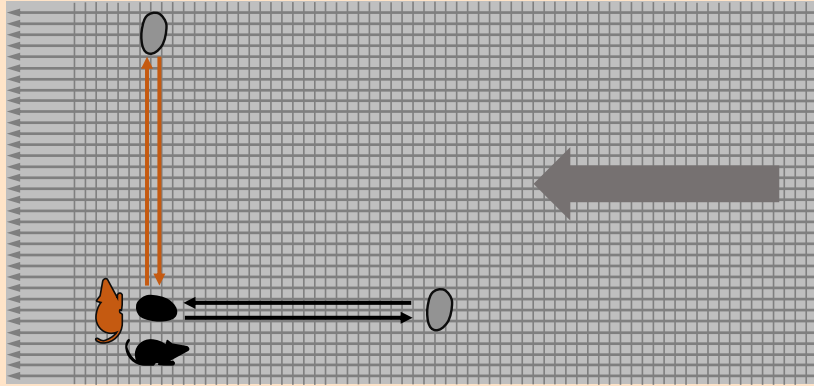
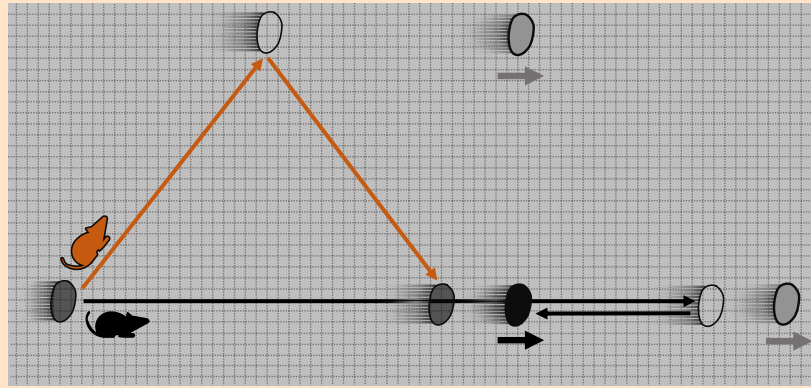


Figure 1.4: **Rats on a treadmill.** A diagram, showing a 3d view of a treadmill platform, with two rats and three hanging rocks, with arrows representing the rats moving from a starting rock to the other two equally spaced rocks, with the paths perpendicular to each other.

If the platform is at rest, they will return to the starting rock simultaneously. But if the treadmill is on and the platform is moving, the rats will also have to work against the platform's movement to reach the rocks. This will lead to different distances the rats have to travel relative to the platform, and as a result, the rats return to the starting rock at different times. Figure (1.5) shows what the direction of movement of the rats and rocks looks like in each of the reference frames when the platform is moving.



(a) Room's reference frame.



(b) Platform's reference frame.

Figure 1.5: **Different reference frames of rats on a treadmill.** Rats running on a moving platform of a treadmill from two different reference frames, with the paths of the rats represented as arrows.

From the room's perspective, the platform is moving backwards. The rat moving in the direction of the treadmill's motion will experience impedance to its movement from the backwards pull of the platform. But after it turns around, it will have a boost from the platform to return to the starting rock. Meanwhile, the sideways-moving rat will have to move sideways and balance out the platform's backwards pull to keep its movement in the direction of the hanging rocks. This gives different path lengths depending on the reference frame, as shown in Figure (1.5). The rats in the platform's reference frame move with the same speed for all the paths, represented by the arrows.

Whereas, in the room's reference frame, we have a slower constant speed of the sideways-moving rat and two different speeds for the forward and backward paths of the other rat. This leads to different times for the rats to return to the starting rock.

The rats here use the medium of the treadmill's platform to move, and similarly, it was thought that light needed a medium, called the aether, to travel through space. If there were such a medium, we could test how fast it moved relative to Earth by measuring the difference in the return times of light emitted in two perpendicular directions, just as in the treadmill example. So, next, we will look at the notion of a universal medium, referred to as the aether, and whether we can have the same frame switching for light traveling in two different frames of reference. As we did in this section.

1.4 The Aether

In the theory of electromagnetism from the late 1800s, it was believed that light was a wave and, therefore, would need a medium that filled the vacuum of space for it to travel through, called "the luminiferous aether", and that light traveled at a constant speed relative to this aether, like how the rats in the previous section moved at a constant speed relative to the medium of the treadmill's platform. It was from the theory of electromagnetism and the experiments performed from it that showed that light should have a constant speed, at least relative to this aether, but it was yet to be known that light would actually have a constant speed relative to all things.

If the aether existed, the Earth would be moving through it, as shown in Figure (1.6).

So, an experimental setup by Michelson and Morley, shown in Figure (1.7), called an interferometer, was devised to measure Earth's movement through the aether [1] by measuring how it affected the return times of light emitted in different directions, as observed in Earth's reference frame. It did this by splitting a single light beam into two perpendicular paths, which were

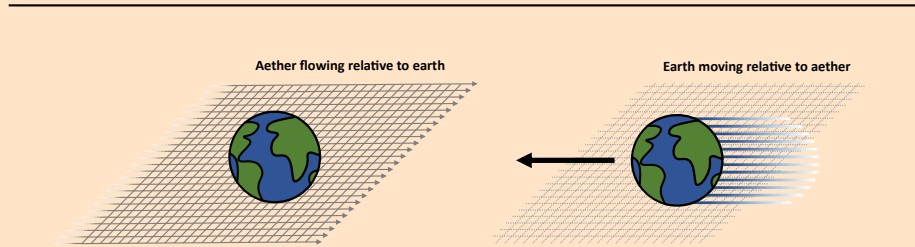


Figure 1.6: **The earth and the aether.** A diagram showing the proposed aether's and Earth's movement relative to each other.

then reflected back to be recombined and sent toward a light detector. By rotating a platform that the setup is on, the two light paths could be aligned, one parallel and the other perpendicular to the Earth's motion through the presumed aether. They reasoned that if the speed of light were constant relative to the proposed aether, as we had for the rats in the previous section, the split light beams would recombine at different times.

From the previous section's system of rats on a treadmill, we can see that the treadmill's platform is analogous to the aether, and the rats are analogous to the light. Both have the room where these experiments are done as the other inertial reference frame.

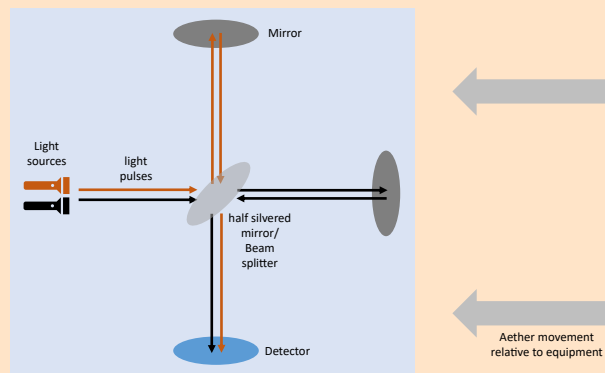


Figure 1.7: **Experiment to detect the aether.** A diagram, showing the Michelson-Morley experiment, we can treat the paths between the beam splitter and the two mirrors to be analogous to the paths in the previous section of the rat and treadmill system.

However, when Michelson and Morley performed the experiment, they found

no difference in travel time to the detector for both paths, indicating that there was no difference in the speed of light in any direction in Earth's reference frame. Hence, light's speed is not dependent on the supposed aether. This null result seriously discredited the aether theories. It led to the proposal by Einstein in 1905 that the speed of light (in a vacuum) is a universal constant, independent of the motion of the observer or the source. To have a universal speed of light (in a vacuum), it will require us to change our ideas about how time and position are perceived by different observers.

1.5 Speed of Light

Experiments conducted in a vacuum showed that light does not just travel at a constant velocity relative to the proposed medium (the aether), but actually travels at a constant velocity relative to all reference frames. The observed speed of light is independent of how fast the source and receiver of that light are moving within the frame, as shown with a moving truck's headlights in Figure (1.8). Light moves more slowly in materials such as glass, but this is due to interactions between the light and the material, which impede its movement.

Light itself moves extremely fast compared to any other everyday speeds we are used to (roughly 300,000,000 meters per second; it can travel the world's diameter in the blink of an eye). This is why we do not notice any delay in the light from the things we see in everyday life, and we assume everything is located where we are currently seeing it, though we should remember that there is this delay. For example, the light we currently see from the sun was emitted 8 minutes ago for it to reach us now, and the further an object is located from us, the further back in time we are currently seeing it because of this delay.

When we look at the same truck setup as in Figure (1.1), but now have it in a vacuum, where the cannon firing a cannonball is swapped for headlights emitting light, we will find the same speed of light when measured relative

to the truck or the road, but how can this be true?

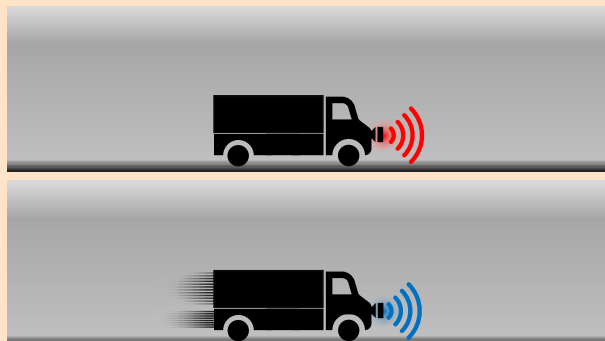


Figure 1.8: **Emitted light in different frames.** A diagram, showing light emitted from a truck in two different reference frames, with the emitted light having the same speed in each frame, though with different frequencies and energies due to the Doppler effect, which will be explained later.

For this to be true, we need a new way of thinking about the addition of velocities. This is because the velocities of objects must be added in a way that is consistent with the requirement that the speed of light is constant but also gives the approximate classical addition at speeds of objects at much less than the speed of light, like how we observed in the situation with the cannon and truck, shown in Figure (1.1). Since the speed of light depends only on the units of time and position, the only way to correct for this is to have the measured positions and times of objects transform differently when switching between reference frames than the classical way. This is what we will talk about next. For the curious, it was the experiment by Ole Rømer ([link](#)) that showed that light travels at a finite speed rather than being instantaneously emitted and received.

1.6 Position and Time

1.6.1 Time Dilation

Consider a simple clock, as shown in Figure (1.9), made of a light pulse moving back and forth vertically between two mirrors on a moving truck, one on the truck's floor and the other on the roof directly above it. Each tick of the clock is defined as the time it takes for the light to travel from one mirror to the other and back. For an observer in the truck, they will see the light go straight up and down between the mirrors, but an observer stationary relative to the road will see not just the light traveling up and down but also with the movement of the truck to allow it to travel back and forth between the mirrors.

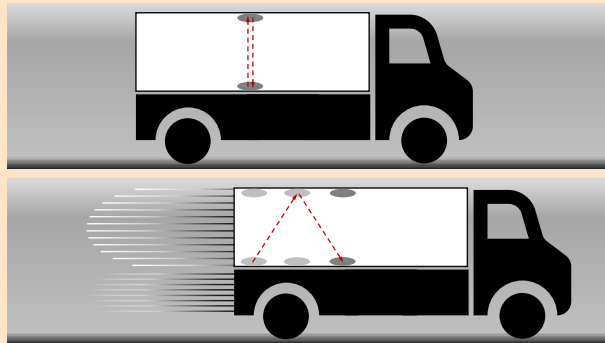


Figure 1.9: **Light clock on a truck.** A diagram, showing a clock consisting of light being reflected between the floor and roof of a truck to keep time, in the truck's frame (top), and in the road's frame with the truck moving (bottom).

Given that the distance light travels is longer in the frame with the truck moving, and that from the previous sections we know that the speed of light is the same in both reference frames, to keep the speed of light invariant, the clock itself must tick more slowly in moving objects.

How much slower time is passing in the moving truck relative to the road observer can be determined from the ratio of the lengths of the paths in each frame, as shown in Figure (1.9), which is the same ratio as the time between

the ticks.

Importantly, this time dilation phenomenon isn't limited to light clocks; it applies universally to all types of clocks. If you were to play a movie on the truck, it would take longer to play through from start to finish for an observer watching it from the road's reference frame. An observer on the road and in the truck experience time at different rates. This is not just an illusion; it is a physical reality.

The closer the truck goes to the speed of light, the more significant the difference in how fast the road frame's time flows relative to the truck frame's time. This can be seen as an increasingly longer horizontal stretch in the path of the light from the perspective of someone standing still on the road, with the time between each tick approaching infinity as the truck approaches the speed of light.

You may ask about the possibility that the distance between the floor and roof of the truck becoming smaller the faster the truck is moving relative to the road, to allow for the light clock to tick at the same speed, and to get rid of the need for the time dilation, but that would actually lead to contradictions which will be shown later in figure (1.12).

Now, if we instead had two trucks moving at the same speed relative to the road but in opposite directions. Each truck would see its light moving vertically up and down with the other truck's light moving a longer diagonal path. Hence, each truck perceives the other truck's clock ticking more slowly. An observer in the road's reference frame would see both trucks move at the same speed and, hence, their diagonal paths having the same length, giving the same passage of time between ticks for both trucks. That is, the slowing of time for moving objects is the same regardless of their relative direction of motion; only the relative speed determines how far the light travels to create each tick of the clock. This concept is confusing and difficult to visualize, but it will get easier with practice.

For now, though, we will continue building up the concepts of special relativity.

1.6.2 Simultaneity

Imagine a truck in its rest frame, with a light bulb in the middle and mirrors on the front and back walls. If the light bulb emits a pulse of light, the light will travel from the center of the truck to reach the mirrors simultaneously, and also back to reach the light bulb simultaneously. An observer on the road watching the truck drive past will see the light bulb simultaneously emit light in both directions and simultaneously return to the bulb. However, the light reflects off each wall's mirrors at different times in the road's frame to do this, as shown in Figure (1.10). This is because the speed of light is the same for both directions, but the truck is moving, meaning the back of the truck is moving towards where the bulb was when the light pulse was emitted, making the distance traveled shorter. The light travels a longer distance to the front wall because the wall has moved further away from when it was emitted. Because of this, the observer on the road will see the light hit the mirrors at different times.

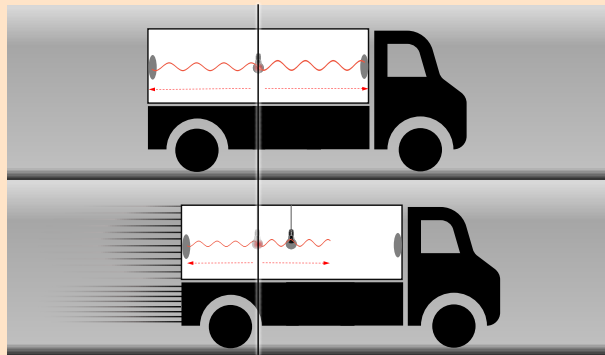


Figure 1.10: **Simultaneity of events in different frames.** A diagram showing how two events of light reaching the two walls of the truck are simultaneous in one frame but occurring at different times in another, due to the truck's movement in the second frame and the speed of light remaining the same.

Consequently, the times at which events occur, such as when the light reaches either mirror, differ for observers in different reference frames. There is no one true order of events. For example, an observer in a faster-moving vehicle

moving in the opposite direction would see the light reach the front wall first. However, for all observers, the light will return to the central light bulb simultaneously. If two events occur at the same position simultaneously, then they occur simultaneously in all frames of reference. This means the light returns to the center of the light bulb simultaneously in each frame. A note here is that we are talking about when the events **actually** happen in each frame, not when the light from those events reaches the observer, which is later due to the time it takes the light from an event to travel to the observer for them to see it. Also, I have left out any mention of the length contraction of the truck, which will be introduced in the next section, and why it must be equal in both directions from the lightbulb for the all the light to return to the light bulb simultaneously.

1.6.3 Length Contraction

Consider a truck, in its rest frame, carrying a square container. If a light bulb at the square container's center emits light in the two horizontal directions and the two vertical directions, it will bounce off the mirrored sides of the truck and return to the center simultaneously. We require that, in the moving frame, they also return to the center simultaneously, as multiple events that occur at a single point simultaneously in one frame, must occur simultaneously in all other frames. The light reflects off the walls at different locations, so the order of reflections in time will depend on the observer, as described in the previous section. The time between the light being emitted and absorbed from the light bulb is the dilated time, as described in section (1.6.1).

To achieve this simultaneity in the return of the light to the bulb, the total length of the path of light in each of the four directions must be the same, as light's speed is the same in all directions. We can determine the length of the up and down path from the time dilation section (1.6.1), and this is the length the path needs to be in the horizontal directions as well. The paths can only

have this length if the truck is contracted in the direction of its motion, to allow for the light to be reflected from the walls at the required positions for these path lengths. The amount of contraction is the same in both directions. The exact amount of the contraction can be worked out geometrically, and it turns out that the ratio of the increase in the amount of time that passes before the light is reabsorbed is inversely proportional to the amount the truck is contracted in the direction of its movement. That is, if in a moving frame the time is dilated so that the light clock takes twice as long between ticks, then the length of the moving truck is half its length at rest. This is shown in the following figure.

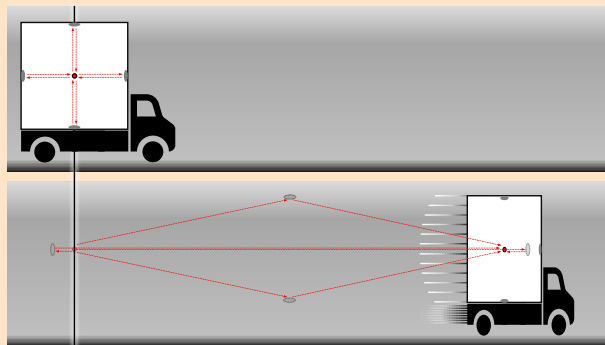


Figure 1.11: **Length contraction of a moving truck.** A diagram, showing a truck with a square container in its rest frame (top), emitting light from a central bulb in the four directions, with all light being reflected by the mirrored sides back to the center, as shown in red. In the second frame, with the truck now moving (bottom), we see its length contracted to allow all paths to have the same length.

We now know that moving objects must shrink in the direction they move to allow for a consistent speed of light in each frame. But this shrinking is not noticeable when objects move much slower than the speed of light. If you are unsatisfied with this, the next section will explain the need for length contraction in another way using three accelerating cars. But first, we will look at the reason we did not consider any change in the sizes of moving objects in the directions perpendicular to an object's movement.

If we consider the possibility that a moving object contracts in the direction perpendicular to its motion, we find that this leads to contradictions, as shown in figure (1.12). As it allows for the possibility that a moving object's size in the perpendicular direction is smaller than a container (that is moving relative to the object) when observed in the container's rest frame, but larger than its container in the object's frame. This is impossible, as this would break the container just because you decided to view the object from a different reference frame.

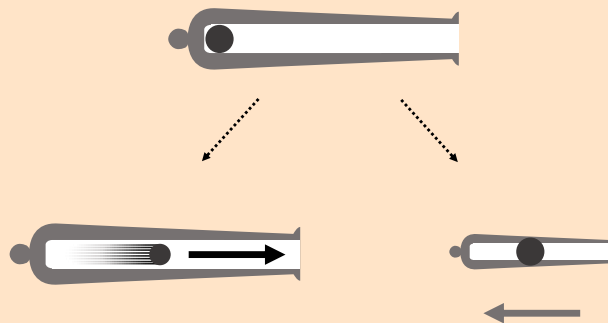


Figure 1.12: **Paradox of height and width contraction.** A diagram showing why there must not be a contraction in the perpendicular direction to the frame's relative motion. The top Figure shows a ball and cannon at rest. The bottom figures show the cannonball being fired in the frame of the cannon (left), where the moving ball has a contracted width, and (right) in the frame of the ball, where the cannon is now moving and has a contracted width, with the ball at rest being the same size. Both frames would contradict each other if there were a change in the width of moving objects, as the walls of the cannon and the surface of the ball would overlap in one frame and have a gap between them in the other. So, we require no size change perpendicular to the object's movement.

In the previous section, in the frame with the truck moving, we saw that the emitted light reaches the top and bottom of the square container at the same time. But the times it reaches the left and right walls are not simultaneous, and it is this difference in timings that allows the truck to contract in the direction of movement without the contradictions that arise when we consider contraction in the perpendicular direction to the movement.

Another Illustration of Length Contraction

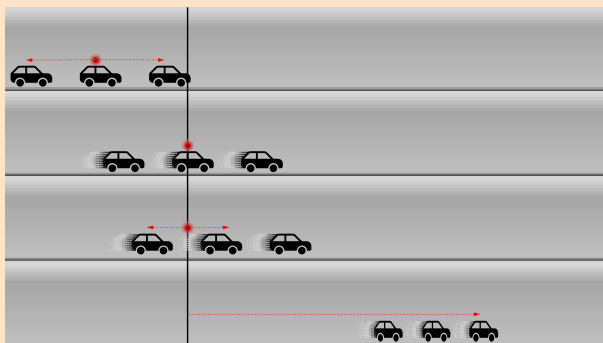


Figure 1.13: **Length contraction between three accelerating cars.** A diagram, illustrating an experiment with three cars initially at rest and equally spaced on a road. The middle car emits light that simultaneously reaches the front and back cars, triggering all cars to accelerate for a predetermined time. Now, another light is emitted from the middle car after the acceleration has finished. The road observer sees the second light pulse reach the back car first, as it moves towards the point where the light was emitted from. This causes it to begin accelerating before the front car. This results in the cars being closer together after acceleration, demonstrating Lorentz length contraction. However, to the car observer, the distance between the cars remains unchanged from the initial distances.

Suppose we have three cars, equally spaced, and at rest on the road, as shown in Figure (1.13). If the middle car emits a pulse of light, it will reach the front and back car simultaneously for the road observer and the car's observers. When it reaches them both, all the cars accelerate for a predetermined fixed amount of time to the same speed. Following this acceleration period, the cars have maintained equal spacing relative to both the car observers and the observer on the road. As for all observers, the light reached the front and back cars at the same time.

Now, suppose we do this a second time, with the middle car releasing another pulse. In this case, within the frame of the observers in the cars, we again see the front and back drivers receive the pulses simultaneously, meaning that they will start accelerating at the same time. After the acceleration has finished, the distances between the cars will be the same for the car observers.

But for the road observer, they see the back car receive the signal first, as that car is moving towards the point where the light pulse was emitted, and the front car is moving away from it. This would mean the back car would accelerate first, getting closer to the front car before the front car begins to accelerate to that same final speed that the back car has already reached. This results in the cars being closer together after the acceleration. The observers in the car and the road observer disagree about what the distances between the cars are. This contraction of length between cars, in the road's frame in which the cars are moving, is called Lorentz length contraction, which means that objects that are moving faster become shorter, and the distances between the objects also become shorter.

1.7 Doppler Effect

If we have a light source at rest, emitting circular pulses of light with equal time intervals between each pulse, we will get concentric circular pulses in this frame. But if we move to a frame in which the source is moving to the right, each circular pulse is emitted from a different position. Due to the source moving, each pulse will be emitted closer to the right-hand side of the previous pulses, creating a bunching of the pulses (increase in frequency) in the direction of movement and a spreading out (decrease in frequency) in the opposite direction. This also happens in the classical version of the Doppler effect. For example, you will notice that an ambulance or police car sounds different when driving towards or away from you, due to the bunching up and spreading of sound waves in the direction and opposite direction of the moving vehicle. However, in special relativity, we must also account for the time dilation of the pulses, which will give a longer time between each subsequent pulse. This is due to objects moving relative to an observer, and therefore having their perceived time moving more slowly. This is called the second-order Doppler effect, most notably seen in optical clocks. It has a decreasing effect on the frequency in all directions. However, in the direction

of the source's movement, this is outweighed by the frequency increase from the previous bunching-up effect. Since the energy of the light is proportional to the frequency, it is also increased in the direction of motion of the source due to the Doppler effects and decreased in the opposite direction.

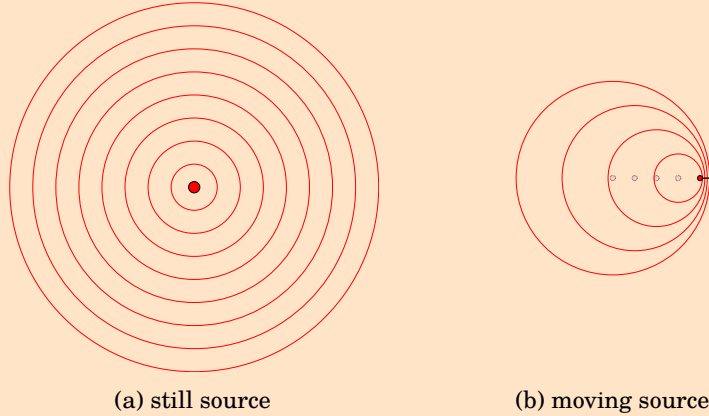


Figure 1.14: **Spherical waves from a moving source.** A diagram showing (left) a central source at rest emitting several circular pulses of light with equal time between each pulse, (right) the same source in a frame where it is now moving and emitting circular pulses of light, but each subsequent pulse is emitted from a different position as the source is moving, marked by a faded dot.

One thing not mentioned yet in this picture is that the light is also affected by a phenomenon called aberration, which changes the angular distribution of the light at each part of the spherical pulses. This will be explained in the next section.

1.8 Aberration

Here, we will talk about what the Doppler effect in the previous section has yet to show, which is the effect on the angular distribution of the light in each part of the spherical pulses. We will show that there is a higher concentration of light in the direction of the source's movement, as shown in Figure (1.15).

With the help of the length contraction section (1.6.3), Consider a truck with a spherically mirrored container, with a central bulb that emits light in all directions. In the rest frame, all light reaches the spherical wall simultaneously and returns to the center bulb simultaneously. Then, in the moving frame, the spherical container's length contracts, and the light moves at the same speed, reflecting off the same parts of the walls but at different times, and then returns to the center bulb simultaneously. For this to be true, the light must be aberrated (have their direction of propagation changed), as shown in the diagram, to allow for the simultaneous return to the bulb.

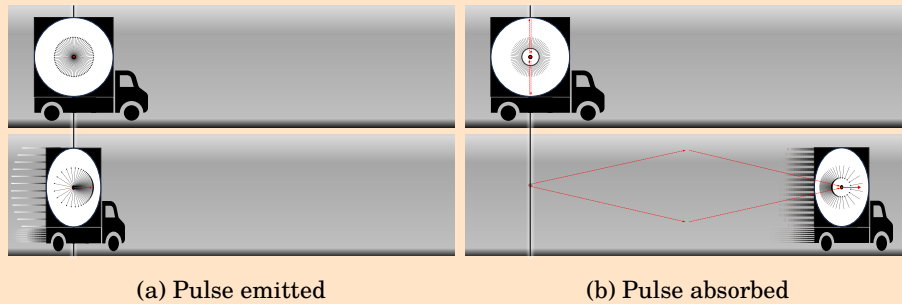


Figure 1.15: **Aberration of light on a truck.** Two diagrams of a truck with a spherical container in two different frames, a light pulse is emitted from the center of the container (left) and reflected back by the mirrored wall to be absorbed in the center again (right), the truck in its rest frame (top of both figures) shows an evenly distributed outward light pulse, but in the frame with the moving truck (bottom of both figures) the emitted light is now more concentrated in the direction of the truck's movement, which is required to allow the light to be reflected by the same part of the container and returned to the center of the truck simultaneously.

From the diagrams, you can see how the direction of light's propagation is changed when emitted or absorbed by a moving source relative to its rest frame. This is called relativistic aberration. The faster the source is moving, the more the direction of each part of the light pulse is aberrated.

If the speed of a source were to approach the speed of light, all emitted light's propagation direction would approach the direction of movement of

the source. If the source could theoretically reach the speed of light, all light would be emitted in the direction of the source, but also move at the same speed. That is, the source would move with the emitted light, and it would not leave the vicinity of the source. The rate at which it emits the light would also tend to zero. If a photon theoretically had mass and its influence on the gravitational force moved at the speed of light, then we would not be able to feel any gravitational effects from this theoretical photon with mass outside its vicinity. This is due to all its gravitational field being propagated in the direction of its movement and at the same speed as the photon.

Now, let us look at how the aberration effect is used when astronomers map the stars in the sky.

1.9 Aberration When Observing the Stars

Earth's view of the universe is distorted differently by the aberration effect, depending on which direction the Earth is currently moving in its orbit around the Sun. This needs to be taken into account when making observations with telescopes, so that we get the locations of stars that agree with measurements when the Earth is moving in different directions.

To get the locations of the stars that are consistent throughout the year, astronomers can calculate the direction of the received light in the rest frame of the sun instead of the Earth, as the sun's movement is more consistent. This is shown in the following diagram.

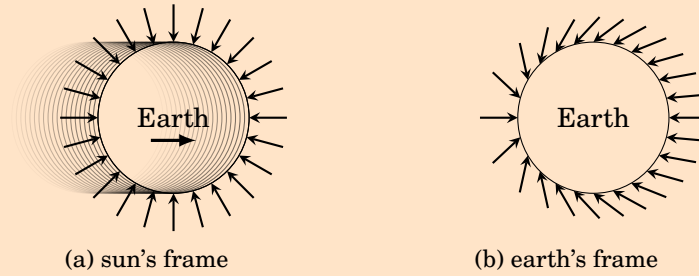


Figure 1.16: **Aberration of observed starlight.** A diagram showing light from distant stars being observed on earth, with the light being received in the sun's frame, with the earth moving (left), and the other diagram (right) showing the same light being received, but now in the earth's rest frame, with the light's direction now more in the direction opposite to earth's movement compared to the sun's frame. The arrows show the direction of the light from the stars.

The sun is also moving relative to the Milky Way galaxy it is orbiting. Therefore, you could go a step further and always calculate the light's direction with respect to the center of the galaxy.

Now, we will look at the combination of the Doppler effect and aberration in the next section.

1.10 Relativistic Beaming

When the Doppler effect and relativistic aberration are combined, it is known as relativistic beaming. If we have a source in its rest frame emitting a spherical pulse of light with evenly distributed angles. Then in a second frame with the source now moving, the Doppler effect has the spherical pulse's wavelength bunched up in the direction of movement (giving a change in the color of the light), with the aberrational effect also giving the angular spread of the light at a higher concentration in the direction of the movement of the source, as shown in figures (1.17).

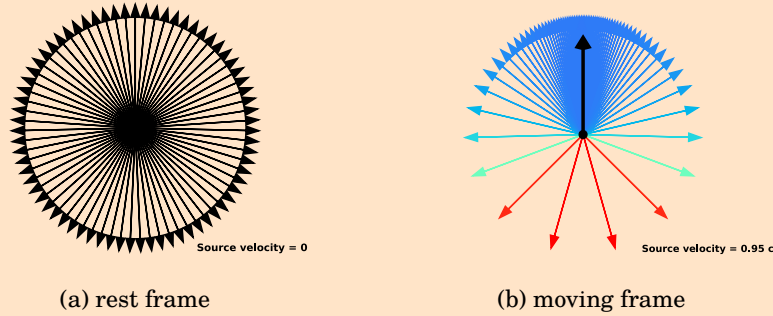


Figure 1.17: **Relativistic beaming of emitted light.** An outward light pulse from a source in its rest and moving frame.

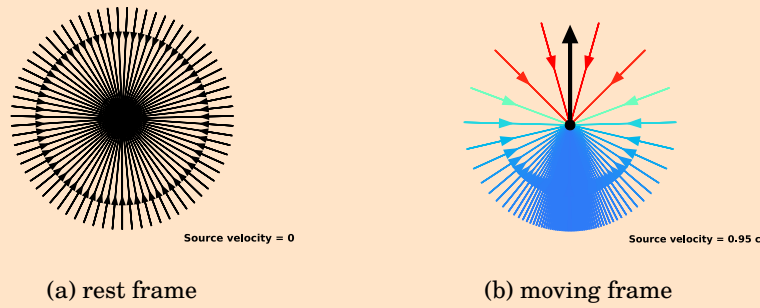


Figure 1.18: **Relativistic beaming of received light.** An inward light pulse to a receiver in its rest and moving frame.

In astrophysics, they need to account for beaming when observing distant stars that are moving relative to us, so they can accurately calculate their brightness and the spectrum of light they emit, among other things. We also need to consider that the position we are currently seeing the stars in is their past position, due to the delay in light reaching us on Earth, which is referred to as the retarded position. We will talk about this next.

1.11 A Source's Retarded Field of Light

We can make a vector field plot, which shows the position and direction of light in space as arrows. So if we have a source in its rest frame that

continually emits light pulses with a constant time interval between them, we get what is shown in Figure (1.19a). If we wanted to know what this light vector field looked like in a frame where the source is moving, we would need to account for the fact that the position from which the source emitted each pulse, is its past position (retarded position). We also have the dilated time between pulses in this moving frame, so using these and the beaming effect from the previous section, we get the light's vector field as shown in Figure (1.19b). From this, we can work out the distribution and, hence, the concentration of the light for any moving source relative to when it is at rest. If you look closely at the diagrams, you will see the aberrated rings of all the pulses, all with different points from which they were emitted, with them being more compact in the direction of the source's movement.

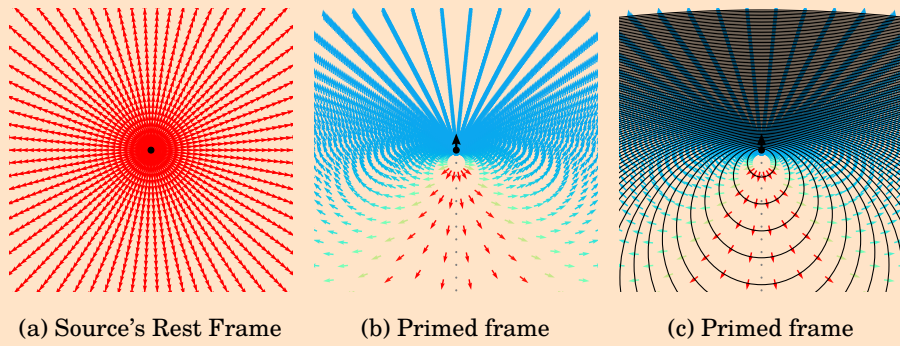


Figure 1.19: Vector field of light pulses from a source. A diagram, showing multiple spherically symmetrical pulses of light from a source at rest, fig (a), and in a corresponding inertial frame where the source is now moving, fig (b), the colors show the Doppler effect on the light, and the gray dots show the past positions of source when it had emitted each pulse, there is also a diagram showing the circles of each pulse, fig (c), This shows the spherical waves being closer together in the direction of movement.

In particle physics, the photon is the particle that composes light. It is the carrier of the electromagnetic force between charges. So, how the light's propagation transforms when switching between frames is necessary to understand electromagnetism, and how the magnetic field comes about for moving sources. The diagram of the moving source shows where the source

and the light currently are. But, if you were an observer at a particular position in the vector field, you would see the source at its retarded position and not where it currently is, as the light has to travel to you before you see it. This will be used in the next section to show that you can perceive things moving faster than light, even if they are not.

1.12 Velocity

We looked at how velocities at the speed of light change between different frames of reference, that is, how their directions of propagation are rotated (aberrated). However, for objects moving at velocities slower than that of light, their speed will change in addition to the direction being transformed when switching between frames. This is because we need the relativistic transform to approach the classical transform as we consider lower and lower speeds of objects. To understand the general velocity transformation, some mathematical detail is needed, so we will brush over this for now and instead look at how the apparent velocities of objects can be more than the speed of light, even if they are actually moving more slowly.

1.13 Apparent versus Actual Velocity

Imagine that a ball one light year away (the distance it takes light to travel in a year) is fired directly towards you, as shown in Figure (1.20). It will take one year for the first light from when the ball starts moving to reach you. If the ball moves at three-quarters of the speed of light, it will hit you in four-thirds of a year (a year and four months) after it is fired. The last light from the ball will reach you just as the ball hits you. As you perceive it, the time between the first and last light from the moving ball is four months. During those four months, you will see the ball start at its initial position and travel a distance of one light-year. So, to you, the ball appears to have been moving three times faster than light. This is just how it appears to the

observer due to the delay in the light signal, giving the latency in how the system is observed.

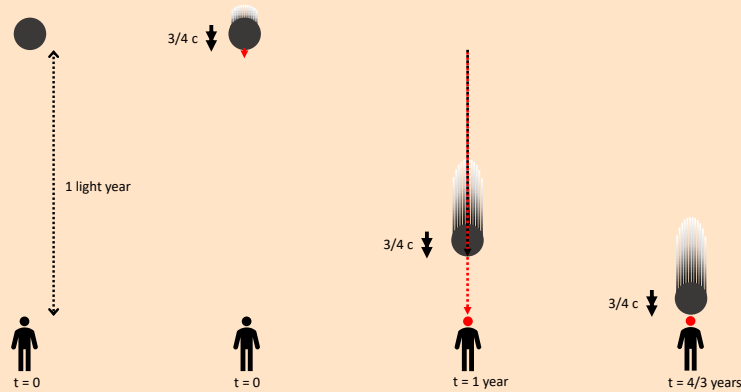


Figure 1.20: **Apparent faster than light movement.** A diagram demonstrating the Apparent speed of a ball vs its actual speed ... the red arrow represents the light being emitted from when the ball starts moving from its position one light year away...

This shows how important it is to account for the delay in light from objects when observing relativistic systems. This view is called the delayed/retarded view, meaning that what we see now are objects in their past positions, and the further away objects are from us, the further into the past we are currently seeing them.

1.14 Summary

A key early concept was that light traveled through a medium called the luminiferous aether. The Michelson-Morley experiment attempted to measure Earth's motion relative to this aether. Surprisingly, no difference in the speed of light was found through this proposed aether in any direction. This conflicted with the intuitive addition of velocities, discredited the aether theory, and led Einstein to propose that light travels at a constant speed in all inertial frames of reference as a fundamental principle.

Special relativity emerged from this insight that light's speed in a vacuum

is constant for all observers, regardless of the light source's speed relative to each observer. This required rethinking the concepts of time intervals and distances between points to accommodate light's fixed speed. It led to the requirement that a clock moving relative to an observer ticks more slowly from the observer's perspective while also being contracted in the direction of its motion. This was shown in Figure (1.9) and Figure (1.11). The faster an object moves relative to an observer, the more its length contracts and its time dilates to the observer.

It also led to the simultaneity of events not being absolute. For example, light emitted from the middle of a moving truck reaches the front and back walls simultaneously for an observer at rest in the truck. But an observer standing still on the road sees the light hit both walls at different times, as shown in Figure (1.10). There is no universal "now" at a distance—observers relate times of events differently. It is only events that occur simultaneously at the same location, that occur simultaneously in all frames of reference.

Classical physics is verified by our day-to-day experience, though this does not reflect the full reality, as relativistic effects only become readily noticeable at speeds close to that of light. With the speed of light being 300,000,000 meters per second, astronomically faster than what we experience in our daily lives.

When it comes to velocities, they combine differently than in classical physics. However, we must ensure that the transforms reduce to the intuitive addition of velocities of classical physics at low speeds while accommodating light's constant rate in all frames of reference. This leads to initially bizarre outcomes, such as **apparent** faster-than-light motion, which arises due to the delay in light signals from objects at a distance. Nevertheless, special relativity gives a truer, more accurate picture of reality.

Due to space-time effects on wave propagation, we also have seen that the emitted light's frequencies shift (Doppler effect) for a moving source, and that light is concentrated in the direction of the source's motion (aberrational effect). Together, these effects both focus light in the direction of moving

sources. This effect is called relativistic beaming.

These effects will all be explained more fully with math in the coming chapters, along with other effects that require the math to fully understand, as well as new energy, mass, and momentum conservation laws. We will derive everything step by step, starting with the transformation of positions and times between reference frames.

References

- [1] Morley Williams Michelson Abraham. “On the Relative Motion of the Earth and the Luminiferous Ether”. In: *American Journal of Science* 34 (1887).

Part II

The Essential Concepts Derived

Here, we will go through the mathematical details of the essential concepts of special relativity, all derived step by step. The chapters in this part teach the basics of special relativity. When taught at university, it typically makes up only a part of one module in the first year and is taught over 2–3 weeks. The topics in this part will be taught during this time.

Chapter 2

Position, Time, and Velocity

Now that we have learned about the concepts of special relativity, we can use them and let the mathematics begin. Starting with classical relativity, which will give us an understanding of how to transform coordinates between reference frames. We can then later compare the special relativistic coordinate transforms to this classical transform to help with our intuition.

What we learn here will be directly used in all the following chapters to derive more foundational concepts of the theory, as everything else is built upon this chapter, which explains how positions and times are transformed between two inertial frames to keep the speed of light constant.

In this chapter, we will begin by looking at how switching between two inertial frames was done classically. Then, we will use two postulates (foundational assumptions) as the basis for special relativity, and by following the reasoning from the previous chapter, we will learn how to switch between frames in special relativity. That is, we will first derive the time dilation and length contraction equations, so that we can then find the full transform of the space and time coordinates between two inertial frames. In doing so, we

will encounter a very useful quantity, which we will denote as γ (gamma). This quantity will be used throughout this book to simplify equations. Lastly, we will look at the transform for infinitesimal (tiny) changes in distances and use this to derive how velocities and accelerations change when switching between two inertial frames. By the end of this chapter, you will understand coordinate transforms between two frames, both conceptually and mathematically. You will also understand the velocity and acceleration transforms, and how to derive them.

2.1 Classical Relativity

Classical (Galilean) relativity was the previous theory of how to transform coordinates to different inertial frames. In many situations in physics, it is a good enough approximation and is still used as the standard.

It was based on two assumptions: **In each frame, you observe the following;**

- The distance between objects is the same.
- Time flows at the same rate in each frame, and therefore, the time that an event occurs is the same in each frame.

These two statements will be the basis for our derivations.

So let us see how the mathematics works for the classical transform of position and time coordinates between two inertial observers' frames, by looking at the coordinates of an event in an initial reference frame with a second inertial frame's coordinate axis moving relative to the initial frame's, illustrated in Figure (2.1):

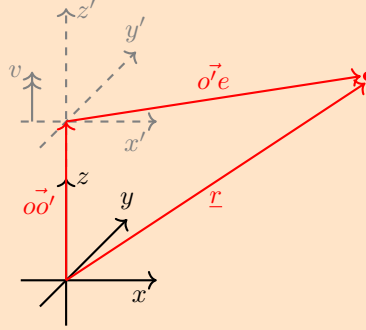


Figure 2.1: **An event in the initial frame with a primed frame's axis moving relative to its axis.** This diagram shows an initial frame with its associated x, y, z coordinate system at rest in this frame, with origin $\underline{o}$. There is also an x', y', z' -coordinate system with origin $\underline{o}'$ associated with the "primed" frame, at position $\underline{o}\underline{o}' = (0, 0, vt)$ in the initial frame, where v is the speed of the primed frame in the z -direction relative to the initial frame, and t is the time in the initial frame. An event $\underline{e}$ in this initial frame is described by a set of spatial and time coordinates $\underline{r} = (x, y, z)$ and t . We choose the time to be zero in both frames when both axes overlapped, so that t is also the time from when they overlapped. We also show $\underline{o}'\underline{e}$, which represents the vector from the primed axis origin to the event in the initial frame.

As shown in the diagram, we are in the initial frame with origin $\underline{o}$ defined by coordinates $(0, 0, 0)$, and the primed frame is moving in the z -direction at velocity v such that the position of the primed origin relative to the initial frame's axis is

$$\underline{o}\underline{o}' = (0, 0, vt) \quad (2.1)$$

at time t from when the origins overlapped. If there is an event at position

$$\underline{r} = (x, y, z) \quad (2.2)$$

the event is described by

$$\underline{e} = (x, y, z, t) \quad (2.3)$$

Then, in the primed frame, we will have the event described by the primed

coordinates that we want to transform into, as

$$\underline{e}' = (x', y', z', t') \quad (2.4)$$

To switch into the primed frame's coordinates, we need to recall from the previous assumptions that distances and times are conserved, and therefore the length and directions of all the vectors between $\underline{o}$, $\underline{o}'$ and $\underline{e}$ are the same in each frame, and time flows at the same rate ($t' = t$), from this knowledge and Figure (2.1) we have the event's coordinates described in the primed frame by its coordinate axis as

$$\underline{r}' = (x', y', z') = \underline{r} - \underline{o}\underline{o}' = (x, y, z - vt) \quad (2.5)$$

This equation gives us the full description of the position and time transform for an event as

$$\begin{aligned} x' &= x \\ y' &= y \\ z' &= z - vt \\ \text{at time: } t' &= t \end{aligned}$$

(2.6)

These are the equations that define classical relativity. We can compare these to the equations of the coordinate transforms in special relativity later. They define the event's position relative to the axis of whichever frame that we are in. The special relativistic equations we will derive later will look similar to these, but with a few extra factors.

We can also write the initial frame's event coordinates in terms of the primed coordinates, by rearranging the transformation equations, giving the inverse transform as

$$\begin{aligned}
 x &= x' \\
 y &= y' \\
 z &= z' + vt'
 \end{aligned}
 \tag{2.7}$$

at time: $t = t'$

So the inverse transformation, going from the primed frame to the original, is just a matter of swapping all primed and initial coordinates and changing the sign on the frame velocity. This can be seen at time $t = 0$ when we have the frame's origins overlapping, as we have the freedom to choose either frame to be the original, as no frame is particularly special, and depending on which one you pick, the relative velocity is either v or $-v$. We will have the same rules in special relativity later on when we want to get the inverse transform.

You might have noticed that the direction of the second frame is in the upward z-direction, rather than to the right like it is in the truck diagrams from the previous chapter. This is to help us later when we use spherical polar coordinates and to help us see the symmetries. But that is a couple of chapters away; for now, we will look at the assumptions of special relativity and their consequences, which we will then apply to the same notion of an event that we have in Figure (2.1).

2.2 Postulates of Special Relativity

For classical relativity, we started with two assumptions. It turns out these assumptions need to be thrown away, as they contradict the experimental evidence that the speed of light is constant regardless of the motion of the observer or light's source. We replace these assumptions with two new ones, called the **postulates of special relativity**:

-
1. The speed of light in a vacuum is the same in all inertial/non-accelerating frames of reference.
 2. The laws of physics and the outcome of experiments are the same in all inertial frames of reference. If an observer in an inertial reference frame were to carry out an experiment, they would find the same result if they had carried it out within a different inertial frame instead.

The first postulate here will be the most important for deriving the transform equations of special relativity. It will be used to show how time is dilated (slowed down) for moving objects. The second postulate has the consequence that if events occur simultaneously at an overlapping position in one frame, then they will also occur simultaneously at an overlapping position in any other inertial frame. We will now use these assumptions, along with the concept of a light clock, to demonstrate length contraction.

2.3 Time Dilation

Time passes more slowly for things moving relative to you; this is called time dilation. To explain why it happens, we will use the simple concept of a light clock. It consists of light bouncing back and forth between two mirrors. It keeps time by measuring when light returns to the first mirror and counting each return as one tick of its clock, as shown in Figure (2.2).

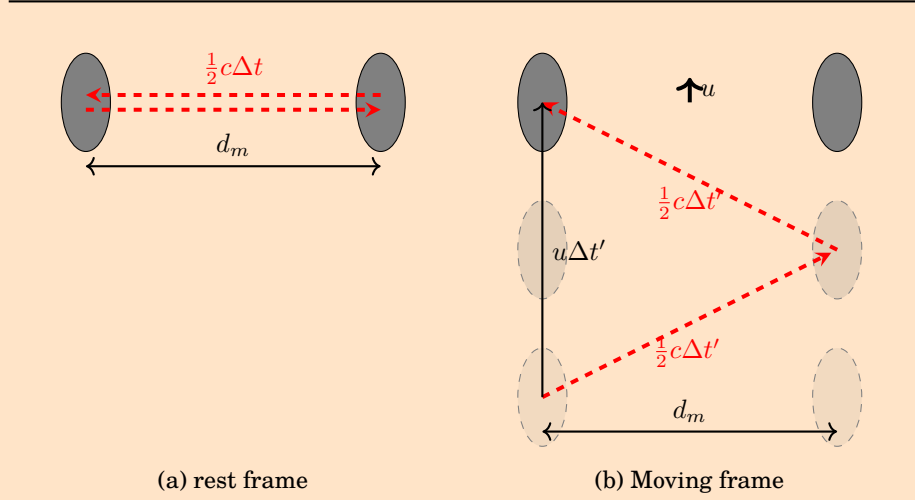


Figure 2.2: Light clock showing time dilation. The light clock keeps time by counting how many round-trips the light has made between the two mirrors separated by a distance $d_m = \frac{1}{2}c\Delta t$, where Δt is the time for the round-trip in the rest frame. The diagram shows the mirror's rest frame (left) and a frame in which the mirror is moving at speed $u = -v$ (right). In this second frame, there is a distance $u\Delta t'$ between where the left mirror reflected the light from and where it returned to it. For this, the light had to travel a total diagonal distance of $c\Delta t'$ to return, where $\Delta t'$ is the time for the return journey in the primed frame.

The light in the clock takes a time Δt in the clock's rest frame to make the return journey. We can see from Figure (2.2a) that in the rest frame, the distance the light travels to the mirror and back in this time is

$$2d_m = c\Delta t. \quad (2.8)$$

The distance the mirrors move during this return journey in the moving frame is

$$u\Delta t' = -v\Delta t', \quad (2.9)$$

where $\Delta t'$ is the time it took light to return within the moving frame, and u is the speed of the light clock, which is in the opposite direction of the movement of the primed frame relative to the rest frame, that is, $u = -v$.

The distance light travels in this time is

$$c\Delta t' \quad (2.10)$$

This is the hypotenuse of the right-angled triangle formed by $u\Delta t' = -v\Delta t'$ and $2d_m$, so we have the square of the hypotenuse equal to the squared sum of the other two lengths

$$c^2\Delta t'^2 = v^2\Delta t'^2 + c^2\Delta t^2 \quad (2.11)$$

rearranging this to find the time it took in the primed frame relative to the rest frame, we get

$$\Delta t' = \gamma\Delta t \quad (2.12)$$

with

$$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}. \quad (2.13)$$

This factor, γ (gamma), is used throughout special relativity to simplify equations. We can see from the diagram and the equations that the time for the return journey is longer in the frame with the clock moving relative to the clock's rest frame,

$$\Delta t' \geq \Delta t \quad (2.14)$$

We take this outcome to mean that the time between two events at the same position in the moving frame is longer than in the rest frame of the clock, because light must travel a longer distance, but at the same speed. Hence, moving systems seem to have time flow more slowly. An example of this is that unstable particles that are moving faster have longer lifespans before they decay.

The earlier time dilation equation (2.12) describes a transformation between

a generic unprimed and primed frame. In later chapters, however, we will be describing specifically the rest frame of a source/particle. We will denote the rest frame's coordinates and properties with a subscript zero, and use unmarked variables for the observer's frame. Consequently, the time transformation shifts from $\Delta t \rightarrow \Delta t'$ to $\Delta t_0 \rightarrow \Delta t$, where Δt_0 explicitly defines the time interval in the rest frame. This gives the updated time dilation equation:

$$\Delta t = \gamma \Delta t_0 \tag{2.15}$$

2.4 Length Contraction

An object's size decreases in the direction of motion compared to when it is at rest; this is called length contraction. To explain why it happens, we will add a second perpendicular light path to the experimental setup from the previous section. So there is now light moving perpendicular as well as parallel the primed frame's movement, as shown in Figure (2.3)

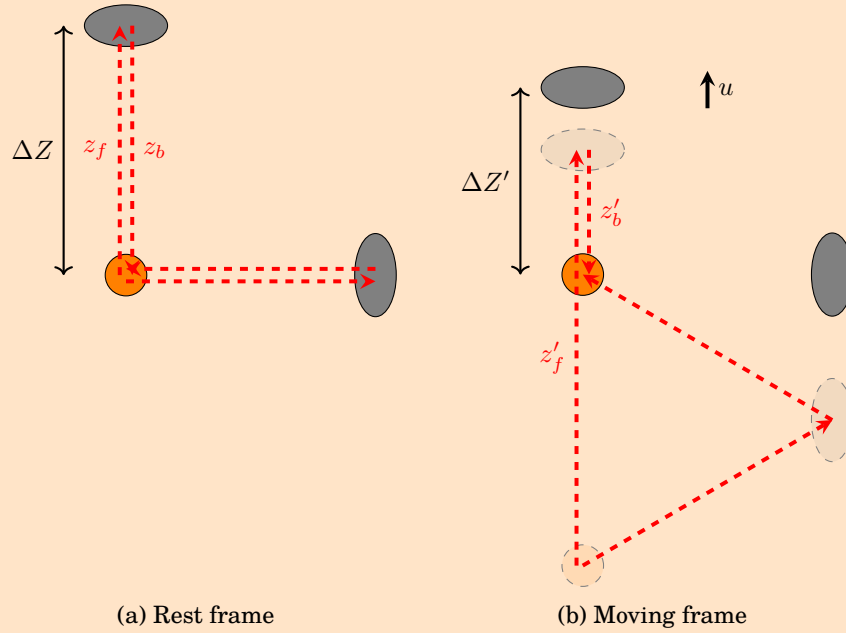


Figure 2.3: Light paths showing the need for length contraction. A central orange bulb emits light in two directions simultaneously; the mirrors then reflect back the light to reach the central bulb simultaneously. The light paths are shown at rest (fig (a)) and in another inertial frame in which the system is moving at speed u (fig (b)), with its length now contracted in the direction of motion to allow for the simultaneous return of light to the bulb. The length between the bulb and front mirror is shown as ΔZ and $\Delta Z'$ in each frame. The paths the light took to the front mirror z_f and the back mirror z_b are shown in the rest frame, with the corresponding paths z'_f and z'_b in the moving frame.

If we have a system as described in Figure (2.3). Then, since the light returns simultaneously in the rest frame, it must also return simultaneously in the second frame. Multiple events that occur at a single point simultaneously in a frame, must then occur simultaneously at a single point in all other frames. This is to ensure the second postulate holds true. The time between the light being emitted and absorbed will be the dilated time that we worked out in the previous section. As the sideways light path in this setup and the light clock's setup from the previous section is the same, hence it takes the same amount of time. In the moving frame, to achieve this simultaneity in

the return of the light to the bulb, the length of the full path of light from the central bulb to the front mirror and back ($2\Delta Z' = z'_f + z'_b$) must be the same as the perpendicular path, due to the speed of light being the same for both paths. We can work out the length of the sideways paths from the time dilation section, $c\Delta t'$, and this is also the total length the other path needs to be. We will find that the paths can only have this length if the distance between the bulb and the front mirror is contracted in the direction of movement. Note that there can not be a contraction in the perpendicular direction to the movement due to the paradox explained in Figure (1.12). In the moving frame, the pulse sent forward has a path length equal to the speed of light multiplied by the time along for the path. Which is also equal to the primed frame's distance between the bulb and mirror plus the distance the mirror has moved away from the point it was emitted in this time, that is

$$z'_f = c\Delta t'_f = \Delta Z' + u\Delta t'_f \quad (2.16)$$

where u is the speed that the front mirror is moving away from the point it was emitted, $\Delta t'_f$ is the time taken to get to the mirror, and $\Delta Z'$ is the distance between the bulb and the mirror in front in the primed frame. After reaching the mirror, the light then must travel back to the bulb a distance

$$z'_b = c\Delta t'_b = \Delta Z' - u\Delta t'_b. \quad (2.17)$$

Where $\Delta t'_b$ is the time taken to return to the bulb from the front mirror, with the source moving towards the reflected light.

These give the two times as

$$\Delta t'_f = \frac{\Delta Z'}{c - u} \quad (2.18)$$

and

$$\Delta t'_b = \frac{\Delta Z'}{c + u}. \quad (2.19)$$

Adding both these times together gives the total dilated time of return as

$$\begin{aligned}
\Delta t' &= \Delta Z' \left(\frac{1}{c+u} + \frac{1}{c-u} \right) \\
&= \frac{\Delta Z'}{c} \frac{2}{1-u^2/c^2} \\
&= 2\gamma^2 \frac{\Delta Z'}{c}
\end{aligned} \tag{2.20}$$

where we used $u = -v$ and the γ factor to simplify the equation. Now, from this and using the time dilation equation (2.12), we have

$$c\Delta t' = c\gamma\Delta t = 2\gamma\Delta Z = 2\gamma^2\Delta Z' \tag{2.21}$$

Finally, this leads to the length from the bulb to the front mirror in the primed frame to be

$$\Delta Z' = \frac{1}{\gamma}\Delta Z. \tag{2.22}$$

This equation shows that for a system moving relative to an observer, the distances for the system in the direction of movement are contracted compared to the rest frame of the object, that is

$$\Delta Z' \leq \Delta Z. \tag{2.23}$$

Objects and the distance between them are therefore shorter in a frame with them moving compared to in the rest frame of the objects.

With this knowledge and the last section, we can now derive the coordinate transform equations.

2.5 Position and Time Transformation

In the last two sections, we have looked at the dilation between two different times at the same location and the contraction between two different positions at the same time, when we switch between frames. Now we will use both of

these concepts to see how the combination of time and position coordinates, known as an event, transforms when we switch frames. For the derivation of the coordinate transformation of an event between two inertial frames, we will start with the same diagram we had in classical relativity in Section (2.1) and apply what we have learned from the time dilation and length contraction sections to maintain consistency with the postulates of special relativity. Our knowledge from this will allow us to derive the coordinate transformation equations, known as the Lorentz transformation.

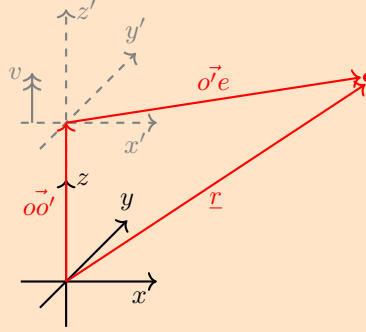


Figure 2.4: An event in an initial frame with a primed frame's axis moving relative to it. This diagram shows an initial frame with its associated x, y, z -coordinate system at rest in this frame, with origin O . There is also an x', y', z' -coordinate system with origin O' associated with the "primed" frame, at position $\vec{OO'} = (0, 0, vt)$ in the initial frame, where v is the speed of the primed frame in the z -direction relative to the initial frame. t is the time in the initial frames. An event e in this initial frame is described by a set of spatial and time coordinates $\underline{r} = (x, y, z)$ and t . We choose the time to be zero in both frames when both axes overlapped, so that t is also the time from when they overlapped. We also show $\vec{O'e}$, which represents the vector from the primed origin to the event in the initial frame.

As shown in the diagram, we are in the initial frame with origin O defined by coordinates $(0, 0, 0)$ and the primed frame is moving in the z -direction at velocity v such that the position of the primed origin in the initial frame is

$$\vec{OO'} = (0, 0, vt) \quad (2.24)$$

at time t from when the origins overlapped. If there is an event described by

$$\underline{e} = (x, y, z, t) \quad (2.25)$$

Then the displacement between the initial frame's origin and the event's position in the initial frame is

$$\underline{r} = (x, y, z) \quad (2.26)$$

and the displacement between the primed frame's origin and the event's coordinates in the initial frame is

$$\vec{o'e} = (x, y, z - vt) \quad (2.27)$$

In the primed frame, we will have the event described by the primed coordinates that we want to transform into, as

$$\underline{e}' = (x', y', z', t') \quad (2.28)$$

with the displacement between the primed frame's origin and the event's position in the primed frame given as

$$\underline{r}' = (x', y', z') \quad (2.29)$$

These are just the event's position coordinates in the primed frame.

The primed frame's axis and coordinates are moving relative to the initial frame, so from the previous length contraction section, we would have the displacement between the primed axis origin and the event in the initial frame $\vec{o'e}$, length contracted, in the z -direction, compared to the primed position coordinates in the primed frame $\underline{r}'$, that is

$$\vec{o'e} = (x, y, z - vt) = (x', y', \frac{z'}{\gamma}) \quad (2.30)$$

The x and y -components are unaffected, with the paradox of if they were affected, explained in Figure (1.12). Rearranging for the z -component, we

have

$$(x', y', z') = \underline{r}' = (x, y, \gamma(z - vt)) \quad (2.31)$$

There is symmetry between the frames: if we were originally in the primed frame, the initial frame would instead be moving at $-v$ in the z -direction relative to the primed frame. We could instead follow the above steps, starting with the negative of the original frame velocity and the primed coordinate, and find that the inverse transformation is the same, except it swaps each frame's coordinates in the transform and uses the negative of the previous frame velocity instead. That is

$$(x, y, z) = \underline{r} = (x', y', \gamma(z' + vt')) \quad (2.32)$$

We can substitute both z -coordinate transforms into each other to find the time transform

$$z = \gamma(\gamma(z - vt) + vt') \quad (2.33)$$

and rearranging gives

$$t' = \gamma \left(\left(\frac{1}{\gamma^2} - 1 \right) \frac{z}{v} + t \right) \quad (2.34)$$

with $1/\gamma^2 = 1 - v^2/c^2$ leading to the primed frames time for the event given in initial frames coordinates as

$$t' = \gamma \left(t - \frac{vz}{c^2} \right) \quad (2.35)$$

Again, if we want the inverse transform from the primed to the initial frame, we can swap the primed and initial frame coordinates and add a minus sign to the frame velocity.

$$t = \gamma \left(t' + \frac{vz'}{c^2} \right) \quad (2.36)$$

We now have the full description of the transformation of the position and time of an event as

$$\begin{aligned}x' &= x \\y' &= y \\z' &= \gamma(z - vt) \\ \text{with the time since the overlap of the coordinate axes:} \\t' &= \gamma\left(t - \frac{vz}{c^2}\right) \\ \text{with } \gamma &= \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}\end{aligned} \tag{2.37}$$

These are the equations that define special relativity. They allow us to find an event's position relative to the axis of the frame that we want to switch into.

In the time transform equation the $\frac{vz}{c^2}$ term is the cause of the simultaneity of events at two different positions in one inertial frame no longer being simultaneous in another inertial frame. The γ factor determines the overall change in how fast time flows at a point relative to the initial frame. In the spatial coordinate transform equations, the vt factor is due to both axes' origins moving away from each other, and the γ factor is the cause of the change of lengths in the direction of the frame's movement.

We can see how points representing the coordinates in an initial frame look like in a primed frame of reference by looking at Figure (2.5).

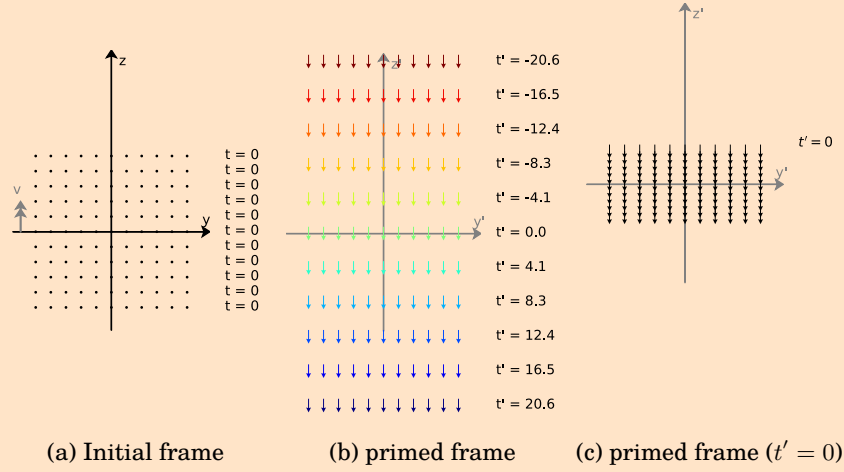


Figure 2.5: **Transformation of coordinates into a primed frame.** A diagram, showing a grid of points, representing the initial frame's coordinates, at time $t = 0$ that are at rest in their frame (left), giving the direct transformation, using the transform equations, into the moving primed frame (middle) where the points are now moving in the negative z -direction with $z' = \gamma z$, the colors and labels show the desynchronization of the primed times $t' = -\frac{v}{c^2}\gamma z$, the next figure (right) shows where these points would be when they are synchronized with the primed origin's time $t' = 0$, which can be worked out to be ($z' = \frac{z}{\gamma}$).

This diagram shows stationary points representing coordinates in the initial frame. These points are transformed to their primed frame positions $z' = \gamma z$ and are now moving at speed $-v$. They are transformed to points that are more spread out in the primed frame. However, the times at which they are located at each of these points are different (they have desynchronized times). Therefore, to find these points, which represent the coordinates in the initial frame, now in the primed frame at the synchronized time $t' = 0$, we must propagate the points to that time, which results in $z' = \frac{z}{\gamma}$, as shown in figure (2.5c).

This figure is to give you a little more intuition on how the coordinates and switching between frames work. We will later see a more natural interpretation of the coordinate transforms when we introduce the notion of retarded time in Chapter (4), which will give us another reason why the time

transformation is the way it is.

2.5.1 Transforms with Small Primed Frame Velocity

If we have the primed frame velocity much smaller than the speed of light ($v \ll c$), then we have $v/c \approx 0$ and therefore $\gamma \approx 1$. Hence, we get back the previous classical transform equations

$$t' \approx t \quad (2.38)$$

$$z' \approx z - vt \quad (2.39)$$

Which is the expected result, as it agrees with the everyday observations we are used to seeing at these speeds. That is, no noticeable difference in clock times or the lengths of moving objects. We can see in Figure (2.6) when the γ -factor's influence becomes significant.

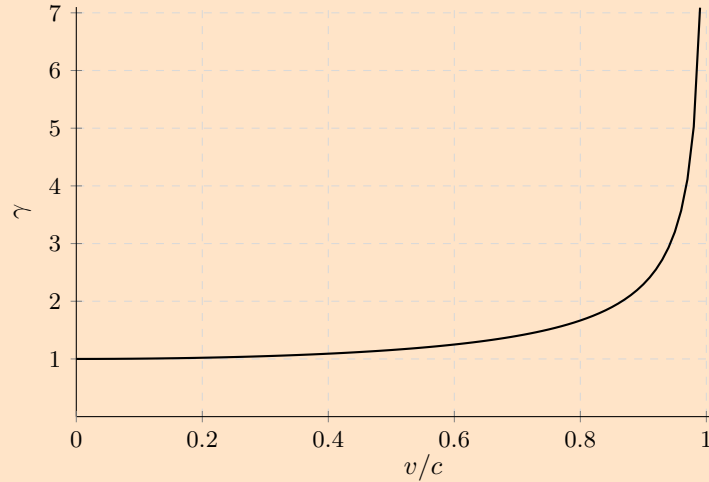


Figure 2.6: **γ -factor.** A diagram, showing the magnitude of γ with increasing ratio of the primed frame's speed to the speed of light, with γ tending to infinity as $v/c \rightarrow 1$.

As can be seen, it is not until the primed frame velocity is approximately a

third of the speed of light that it significantly differs from 1. The speed of light is a speed unattainable to us; the only thing we can bring up close to this speed is particles, and nothing human made can even reach a 1000th the speed of light.

2.5.2 Space and Time Intervals

To get the velocity transform equations, we first need the equations for transforming the displacement between two positions and the interval between two times, then take the limit as the changes go to zero. That is, when we have infinitesimal changes of coordinates. If we want to know how the displacement between two coordinates (x_1, y_1, z_1) and (x_2, y_2, z_2) transforms, we can take the difference of the transforms of both the coordinates as follows

$$\begin{aligned}
x'_2 - x'_1 &= x_2 - x_1 \\
y'_2 - y'_1 &= y_2 - y_1 \\
z'_2 - z'_1 &= \gamma(z_2 - vt_2) - \gamma(z_1 - vt_1) \\
t'_2 - t'_1 &= \gamma\left(t_2 - \frac{v}{c^2}z_2\right) - \gamma\left(t_1 - \frac{v}{c^2}z_1\right)
\end{aligned} \tag{2.40}$$

rearranging and replacing the difference of the first and second of each coordinate with $\Delta_{\underline{r}} = (\Delta x, \Delta y, \Delta z) = (x_2 - x_1, y_2 - y_1, z_2 - z_1)$, giving the displacement transform as

$$\begin{aligned}
\Delta x' &= \Delta x \\
\Delta y' &= \Delta y \\
\Delta z' &= \gamma(\Delta z - v\Delta t) \\
\Delta t' &= \gamma\left(\Delta t - \frac{v}{c^2}\Delta z\right)
\end{aligned} \tag{2.41}$$

This transform has the same structural form as the coordinate transform equations.

From these equations, we can also get the differential quantities when we

take infinitesimal changes, which give the same form again, as

$$\begin{aligned}
dx' &= dx \\
dy' &= dy \\
dz' &= \gamma(dz - vdt) \\
dt' &= \gamma\left(dt - \frac{v}{c^2}dz\right)
\end{aligned}
\tag{2.42}$$

So, when we make an infinitesimally small change in position and/or time within a frame, we know what the corresponding change is in a primed frame. We will use these equations to get the velocity transform equations. The velocity at any instance is the infinitesimal change in position per infinitesimal change in time. These will also be used later to find invariant quantities, quantities that remain the same in each frame. For example, the space-time interval and the energy-momentum relationship in Chapter (3).

2.6 Velocity

The velocity at any instant is the infinitesimal change in position divided by the infinitesimal change in time: $\underline{u} = \frac{d\underline{r}}{dt}$. So using the infinitesimals from Equation (2.42), we get the velocity in each direction as

$$\begin{aligned}
u'_x &= \frac{dx'}{dt'} = \frac{dx}{\gamma\left(dt - \frac{v}{c^2}dz\right)} \\
u'_y &= \frac{dy'}{dt'} = \frac{dy}{\gamma\left(dt - \frac{v}{c^2}dz\right)} \\
u'_z &= \frac{dz'}{dt'} = \frac{\gamma(dz - vdt)}{\gamma\left(dt - \frac{v}{c^2}dz\right)}
\end{aligned}
\tag{2.43}$$

Now, dividing the top and bottom of the fraction on the right-hand side by dt , we can give the equations in their vector form as

$$\underline{u}' = \frac{1}{\gamma \left(1 - \frac{v}{c^2} u_z\right)} \begin{pmatrix} u_x \\ u_y \\ \gamma(u_z - v) \end{pmatrix} \quad (2.44)$$

at position $\underline{r}'$ and time t'

This transform is used to find the velocity of anything in another frame at the corresponding primed position and time. We will put this equation to work in a later section, when we look at how the light emitted by a source has its velocity's direction changed (aberrated) when transforming between frames, and the effect this has on the light's intensity in all directions. Next, we will use this to get the acceleration transform.

Side Note:

For those who are interested, the generalized velocity transform (which we will not derive here) for a frame velocity in any direction, and not just the z-direction, is

$$\underline{u}' = \frac{1}{\gamma} \frac{\underline{u} + \left[\frac{\gamma - 1}{\|\underline{v}\|^2} (\underline{u} \cdot \underline{v}) - \gamma \right] \underline{v}}{1 - \frac{\underline{u} \cdot \underline{v}}{c^2}} \quad (2.45)$$

with $\underline{v}$ being the frame's velocity in any generalized direction.

2.7 Acceleration

To find how the acceleration (known as the 3-acceleration) of a particle transforms for observers in two different frames, we take the differential of the velocity transform in equation (2.44) and use the differentiation rules for two generic functions f and g : $d(gf) = f dg + g df$, and $d[f(g(x))] = dg(x) * df(g(x))$

$$\begin{aligned}
d\underline{u}' &= \frac{1}{\gamma \left(1 - \frac{v}{c^2} u_z\right)} \begin{pmatrix} du_x \\ du_y \\ \gamma du_z \end{pmatrix} + \frac{\gamma \frac{v}{c^2} du_z}{\gamma^2 \left(1 - \frac{v}{c^2} u_z\right)^2} \begin{pmatrix} u_x \\ u_y \\ \gamma (u_z - v) \end{pmatrix} \\
&= \dots \\
&= \frac{1}{\gamma \left(1 - \frac{v}{c^2} u_z\right)^2} \begin{pmatrix} du_x + \frac{v}{c^2} (u_x du_z - u_z du_x) \\ du_y + \frac{v}{c^2} (u_y du_z - u_z du_y) \\ \frac{1}{\gamma} du_z \end{pmatrix}
\end{aligned} \tag{2.46}$$

Now, divide by dt' as we did for the velocity in the previous section, and we get the acceleration transform as

$$\begin{aligned}
\underline{a}' &= \frac{1}{\gamma^2 \left(1 - \frac{v}{c^2} u_z\right)^3} \begin{pmatrix} a_x + \frac{v}{c^2} (u_x a_z - u_z a_x) \\ a_y + \frac{v}{c^2} (u_y a_z - u_z a_y) \\ \frac{1}{\gamma} a_z \end{pmatrix} \\
&\text{at position } \underline{r}' \text{ and time } t'
\end{aligned} \tag{2.47}$$

So if we know how a particle accelerates in one frame, we will know its corresponding acceleration in another frame at the corresponding primed time and position.

Side Note:

Again, for those who are interested, the generalized acceleration transform (which we will not derive here) for a frame velocity in any direction and not just the z-direction is

$$\underline{a}' = \frac{1}{\gamma^3 \left(1 - \frac{(\underline{u} \cdot \underline{v})}{c^2}\right)^3} \left[\gamma \left(1 - \frac{(\underline{u} \cdot \underline{v})}{c^2}\right) \underline{a} + \frac{\gamma (\underline{a} \cdot \underline{v})}{c^2} \underline{u} + (1 - \gamma) (\underline{a} \cdot \underline{\hat{v}}) \cdot \underline{\hat{v}} \right] \tag{2.48}$$

at position $\underline{r}'$ and time t' . This equation is referred to as the 3-acceleration transform in other sources.

Now, when it comes to force, it is a bit more confusing. In classical/Galilean physics force can be described as the rate of change of momentum, or equivalently, as mass multiplied by the acceleration. But in special relativity, we define force as the rate of change of momentum alone, since momentum has an additional factor due to relativistic effects compared to the classical version, as explained in the next chapter. This extra term leads to a different formula for the force compared to the mass times the acceleration above.

2.8 The Paradox of Faster than Light Signals

Here, we will look at sending signals faster than the speed of light and see that this leads to the paradox that the time of events can change the order in which they happen in different reference frames. The thought experiment that examines this is called the Tachyonic Antitelephone and is as follows.

Consider two particles at rest in an initial reference frame, an emitting particle and a receiving particle, with the emitting particle located at the origin, and the receiving particle located on the positive z-axis at z_r .

At time 0, the emitting particle at the origin emits a signal at speed u_s , which is received by the receiving particle at position z_r at time $t_r = z_r/u_s$ in this rest frame of the particles.

Now, consider a primed frame moving at speed v relative to the first. To find the emission and received times in the primed frame, we use the time transformation from equations (2.37). Giving the emission time as $t' = 0$ and the time it was received as

$$t'_r = \gamma \left(t_r - \frac{v z_r}{c^2} \right) = \gamma \left(1 - \frac{v u_s}{c^2} \right) t_r \quad (2.49)$$

since γ and t_r are positive, we have that if $v u_s > c^2$, which is possible if $u_s > c$, then the time signal was received in the primed frame can actually end up being negative (i.e., before time of emission), meaning that you could have the receiving particle receive the signal before the emitting particle has sent

it.

This thought experiment, by itself, does not completely rule out speeds faster than the speed of light. The mathematics does allow for it, but the possibility of the order of a cause and effect being interchangeable would contradict our intuition, even if the math itself stays consistent in all reference frames.

From this, many conclude that it should be impossible to send a signal faster than the speed of light. On top of this, there is also a restriction on particles with mass that leads to another barrier to speeds faster than light, which we will see later in section (3.3.7), which discusses specifically why particles with mass can not reach the speed of light, due to them requiring infinite energy to get to that speed.

2.9 Summary

In this chapter, we derived the mathematics for transforming spatial and time coordinates in classical relativity, which later helped us understand the derivation of special relativity. We then introduced the assumptions, known as the postulates of special relativity, which we use to derive how time slows down due to time dilation and how the lengths of moving objects contract. This led to the introduction of γ , which is a factor used throughout relativity to simplify equations. We then had the tools to derive the transformation equations, which in turn allowed us to calculate the velocity and acceleration transforms. Later in Chapter (4), we will take a look at the delayed view of the world an observer sees, due to the delay in light from their surroundings, and this will help us understand why we need the time transformation to be the way it is.

So we have learned how to transform from one reference frame into another, in special relativity. The outline of steps to transform a system of particles and their properties such as position, time, velocity, and acceleration, is as follows. If you have a system of particles in an initial frame, and you want to transform their positions to a new frame, you need to first define where the

origin of this initial axis is. You are free to choose where it is. The primed frame's axis will overlap with this initial frame's axis at your arbitrarily chosen starting time, that is, when you define the time as $t = 0$ (the time of the overlap is also $t' = 0$ in the primed frame). You can now transform the positions into the primed frame. These particles have their primed frame positions at the calculated primed frame times, defined in Equations (2.37). To transform the velocities and acceleration of these particles at their primed frame positions at the corresponding primed times in the primed frame, you can use equations (2.47) and (2.44).

And this is all the steps needed, but an easy mistake to make is to forget that t and t' are the times from the overlap of the frame origins in their respective frames.

This chapter laid the foundations of special relativity, but there is still more to come.

Chapter 3

Quantities That All Inertial Frames Agree On

Some quantities do not change when we transform between inertial reference frames. These frame-independent quantities are referred to as Lorentz invariant quantities.

The simplest quantity that all inertial frames agree on is the speed at which light propagates. When switching from an initial frame to a primed frame, the speed of light does not change, $c' = c$. We can equivalently express this using the light's velocity vector as $\|\underline{c}'\| = \|\underline{c}\|$.

In this chapter, we will see more invariant quantities related to the space-time coordinates (x, y, z, t) and use them to find invariant properties associated with velocity, which will lead to the more abstract conservation relations among momentum, energy, and mass. We will find a new quantity with the units of energy that has a term which approximates to the kinetic energy of a particle at low velocities and another term that suggests particles have another type of energy, which was not previously in classical physics, that depends on the mass of a particle (mass-energy equivalence). This new quantity will lead to a new conservation law of energy. The reason we care

about the concept of energy is that it has a conservation law, which we will learn about later, that can be used as a tool to help us make calculations and model more complex systems.

3.1 The Space-Time Interval

An event in special relativity is defined by its position and time coordinates x , y , z , and t . These are called its space-time coordinates. The changes in the coordinates between two events are denoted as Δx , Δy , Δz , and Δt .

From these, we could define a quantity called the space-time interval ΔS between two events in an inertial reference frame as:

$$\Delta S^2 = (c\Delta t)^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 \quad (3.1)$$

The reason we care about this new quantity is that all inertial frames of reference agree on the space-time interval between two events; it is invariant between frames, $\Delta S = \Delta S'$. The invariance can be shown to be true when we substitute the coordinate transforms from equation (2.41) into the primed version of this last equation:

$$\begin{aligned} \Delta S'^2 &= (c\Delta t')^2 - \Delta x'^2 - \Delta y'^2 - \Delta z'^2 \\ &= c^2\gamma\left(\Delta t - \frac{v}{c^2}\Delta z\right)^2 - \Delta x^2 - \Delta y^2 - \Delta(\gamma(\Delta z - v\Delta t))^2 \\ &= (c\Delta t)^2 - \Delta x^2 - \Delta y^2 - \Delta z^2 \\ &= \Delta S^2. \end{aligned} \quad (3.2)$$

This relationship will come in very useful in the later energy, mass, and momentum sections.

3.2 D'Alembert operator

(This can be seen as a bonus section that may be skipped, as it is not directly relevant to the understanding of any other sections in this text.)

The d'Alembert operator is fundamental in relativistic wave equations; it is invariant in all frames of reference. It is denoted as $\square$, and consists of the second-order partial derivatives with respect to each coordinate. It is defined in an initial frame as

$$\square = \frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \quad (3.3)$$

and in a primed frame as

$$\square' = \frac{1}{c^2} \frac{\partial^2}{\partial t'^2} - \frac{\partial^2}{\partial x'^2} - \frac{\partial^2}{\partial y'^2} - \frac{\partial^2}{\partial z'^2} \quad (3.4)$$

In the following derivation, we will show that it is invariant; this will be done by transforming the partial derivatives between reference frames and showing that the operator retains its form, and that $\square' = \square$.

derivation:

From the transformation equations (2.37), we can get all partial derivatives, which we will use to transform the equations between frames:

$$\begin{array}{llll} \frac{\partial x'}{\partial x} = 1 & \frac{\partial y'}{\partial x} = 0 & \frac{\partial z'}{\partial x} = 0 & \frac{\partial t'}{\partial x} = 0 \\ \frac{\partial x'}{\partial y} = 0 & \frac{\partial y'}{\partial y} = 1 & \frac{\partial z'}{\partial y} = 0 & \frac{\partial t'}{\partial y} = 0 \\ \frac{\partial x'}{\partial z} = 0 & \frac{\partial y'}{\partial z} = 0 & \frac{\partial z'}{\partial z} = \gamma & \frac{\partial t'}{\partial z} = -\frac{\gamma v}{c^2} \\ \frac{\partial x'}{\partial t} = 0 & \frac{\partial y'}{\partial t} = 0 & \frac{\partial z'}{\partial t} = -\gamma v & \frac{\partial t'}{\partial t} = \gamma \end{array} \quad (3.5)$$

To find the partial derivatives with respect to the unprimed coordinates, in terms of the partial derivatives with respect to the primed coordinates, we can use the chain rule. Which for $\frac{\partial}{\partial t}$ is:

$$\frac{\partial}{\partial t} = \frac{\partial t'}{\partial t} \frac{\partial}{\partial t'} + \frac{\partial x'}{\partial t} \frac{\partial}{\partial x'} + \frac{\partial y'}{\partial t} \frac{\partial}{\partial y'} + \frac{\partial z'}{\partial t} \frac{\partial}{\partial z'} \quad (3.6)$$

and for $\frac{\partial}{\partial z}$:

$$\frac{\partial}{\partial z} = \frac{\partial t'}{\partial z} \frac{\partial}{\partial t'} + \frac{\partial x'}{\partial z} \frac{\partial}{\partial x'} + \frac{\partial y'}{\partial z} \frac{\partial}{\partial y'} + \frac{\partial z'}{\partial z} \frac{\partial}{\partial z'} \quad (3.7)$$

Substituting the partial derivatives from above into these chain rule expressions, yields:

$$\begin{aligned} \frac{\partial}{\partial x} &= \frac{\partial}{\partial x'} \\ \frac{\partial}{\partial y} &= \frac{\partial}{\partial y'} \\ \frac{\partial}{\partial z} &= -\frac{\gamma v}{c^2} \frac{\partial}{\partial t'} + \gamma \frac{\partial}{\partial z'} \\ \frac{\partial}{\partial t} &= \gamma \frac{\partial}{\partial t'} - \gamma v \frac{\partial}{\partial z'} \end{aligned} \quad (3.8)$$

Now, we can get the second-order partial derivatives, which are the same as applying the first-order partial derivative twice, which gives

$$\begin{aligned} \frac{\partial^2}{\partial t^2} &= \frac{\partial}{\partial t} \cdot \frac{\partial}{\partial t} = \left(\gamma \frac{\partial}{\partial t'} - \gamma v \frac{\partial}{\partial z'} \right) \left(\gamma \frac{\partial}{\partial t'} - \gamma v \frac{\partial}{\partial z'} \right) \\ &= \gamma^2 \frac{\partial^2}{\partial t'^2} - 2\gamma^2 v \frac{\partial^2}{\partial t' \partial z'} + \gamma^2 v^2 \frac{\partial^2}{\partial z'^2} \end{aligned} \quad (3.9)$$

$$\begin{aligned} \frac{\partial^2}{\partial z^2} &= \left(-\frac{\gamma v}{c^2} \frac{\partial}{\partial t'} + \gamma \frac{\partial}{\partial z'} \right) \left(-\frac{\gamma v}{c^2} \frac{\partial}{\partial t'} + \gamma \frac{\partial}{\partial z'} \right) \\ &= \frac{\gamma^2 v^2}{c^4} \frac{\partial^2}{\partial t'^2} - \frac{2\gamma^2 v}{c^2} \frac{\partial^2}{\partial t' \partial z'} + \gamma^2 \frac{\partial^2}{\partial z'^2} \end{aligned} \quad (3.10)$$

$$\begin{aligned} \frac{\partial^2}{\partial x^2} &= \frac{\partial^2}{\partial x'^2} \\ \frac{\partial^2}{\partial y^2} &= \frac{\partial^2}{\partial y'^2} \end{aligned} \quad (3.11)$$

We can now substitute these transformed second derivatives into equation (3.3) to get the d'Alembert operator.

Let us first do this for just the t and z components $\left(\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial z^2} \right)$:

$$\begin{aligned}
\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial z^2} &= \frac{1}{c^2} \left(\gamma^2 \frac{\partial^2}{\partial t'^2} - 2\gamma^2 v \frac{\partial^2}{\partial t' \partial z'} + \gamma^2 v^2 \frac{\partial^2}{\partial z'^2} \right) \\
&\quad - \left(\frac{\gamma^2 v^2}{c^4} \frac{\partial^2}{\partial t'^2} - \frac{2\gamma^2 v}{c^2} \frac{\partial^2}{\partial t' \partial z'} + \gamma^2 \frac{\partial^2}{\partial z'^2} \right) \\
&= \frac{1}{c^2} \left(\gamma^2 \frac{\partial^2}{\partial t'^2} + \gamma^2 v^2 \frac{\partial^2}{\partial z'^2} \right) - \left(\frac{\gamma^2 v^2}{c^4} \frac{\partial^2}{\partial t'^2} + \gamma^2 \frac{\partial^2}{\partial z'^2} \right) \\
&= \frac{1}{c^2} \frac{\partial^2}{\partial t'^2} - \frac{\partial^2}{\partial z'^2}
\end{aligned} \tag{3.12}$$

Now adding in the x and y components, to get the full d'Alembert operator in both frames:

$$\frac{1}{c^2} \frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2}{\partial t'^2} - \frac{\partial^2}{\partial x'^2} - \frac{\partial^2}{\partial y'^2} - \frac{\partial^2}{\partial z'^2} \tag{3.13}$$

As can be seen in this equation, the left-hand side is the D'Alembert operator in the initial frame's coordinates, and the right-hand side is the D'Alembert operator in the primed frame's coordinates.

So we can conclude that the D'Alembert operator in both frames are equal:

$$\Box' = \Box \tag{3.14}$$

Thus, we have demonstrated that the d'Alembert operator maintains its form under an inertial frame transformation, confirming its invariance. The invariance of this is a direct consequence of the way transformations preserve the spacetime interval.

3.3 Energy-Mass and Momentum Relation

When it comes to the relationships among energy, mass, and momentum, there are many different derivations. But there is no straightforward deriva-

tion that avoids all downsides, such as hidden assumptions. I wanted to avoid a derivation that seems good on the surface but has its flaws hidden, or just ignored. So I decided to at least start with a strong foundation, and point out any assumptions as we derive it. To do this, we will start with a quantity that we know is invariant, and from it get a quantity with the units of energy that is invariant, which we will use to derive the full relationship. We will use some mathematical tricks, such as the Taylor expansion, and make reasonable interpretations of each term in the equation. For instance, there are quantities in the equations we will encounter later that we will interpret as the total energy and the inherent energy a particle at rest has due to its mass. These interpretations do indeed turn out to agree with experiments [8, 7], which is what matters.

So let us begin by deriving the invariant energy quantity.

3.3.1 The Invariant Energy Quantity

Here, we derive the invariant energy quantity of a particle from the spacetime interval.

derivation:

We can see from equation (3.1) that if a particle q is moving in an inertial reference frame, we can define the space-time interval of its coordinates in this frame as

$$\Delta S_q^2 = c^2 \Delta t_q^2 - \Delta x_q^2 - \Delta y_q^2 - \Delta z_q^2 \quad (3.15)$$

We will not change reference frames anywhere in these next sections. And since we will remain in the same reference frame, we will not need to make use of any primed notation ($'$) here.

Now we can divide this space-time interval by the square of the proper time that passes for the particle in this interval Δt_0 (the interval in time that passes in the particle's rest frame). The proper time is the

time that all frames of reference agree on and that keeps the expression invariant. Now, taking the limit as the change in proper time in the interval goes to zero, we get a derivative that is an abstract invariant quantity that we will call the space-time velocity:

$$\left(\frac{dS_q}{dt_0}\right)^2 = \left(c \frac{dt_q}{dt_0}\right)^2 - \left(\frac{dx_q}{dt_0}\right)^2 - \left(\frac{dy_q}{dt_0}\right)^2 - \left(\frac{dz_q}{dt_0}\right)^2 \quad (3.16)$$

From the time dilation equation (2.15) (using $\Delta t_0 = dt_0$ and $\Delta t = dt_q$), we have $dt_0 = dt_q/\gamma_u$ with $\gamma_u = \frac{1}{\sqrt{1 - \frac{u_q^2}{c^2}}}$. This allows us to reduce this space-time velocity to the following:

$$\begin{aligned} \left(\frac{dS_q}{dt_0}\right)^2 &= \gamma_u^2 \left(c \frac{dt_q}{dt_q}\right)^2 - \gamma_u^2 \left(\frac{dx_q}{dt_q}\right)^2 - \gamma_u^2 \left(\frac{dy_q}{dt_q}\right)^2 - \gamma_u^2 \left(\frac{dz_q}{dt_q}\right)^2 \\ &= c^2 \gamma_u^2 \left(1 - \frac{u_{qx}^2 + u_{qy}^2 + u_{qz}^2}{c^2}\right) \\ &= c^2 \gamma_u^2 \left(1 - \frac{u_q^2}{c^2}\right) \\ &= c^2 \end{aligned} \quad (3.17)$$

It can be seen that this space-time velocity is still an invariant quantity as the speed of light is constant in all frames.

Now we could multiply both sides of the third line in equation (3.17) by the invariant quantity $m^2 c^2$, where m is the invariant rest mass of the particle, so that we get

$$\begin{aligned} m^2 c^2 \left(\frac{dS_q}{dt_0}\right)^2 &= m^2 c^4 \gamma_u^2 \left(1 - \frac{u_q^2}{c^2}\right) \\ (mc^2)^2 &= \gamma_u^2 m^2 c^4 - \gamma_u^2 m^2 u_q^2 c^2 \end{aligned} \quad (3.18)$$

where $\left(\frac{dS_q}{dt_0}\right)^2 = c^2$ was substituted in, to get the final equation.

This final equation's terms are all in units of energy squared, with the

invariant energy squared term on the left, which we will tidy up so that we can write it as

$$(mc^2)^2 = (\gamma_u mc^2)^2 - (\gamma_u mu_q c)^2 \quad (3.19)$$

This invariant expression will serve as our basis for deriving everything in the following sections.

3.3.2 Rest Mass-Energy Equivalence

Equation (3.19) shows an invariant quantity with the units of energy squared $(mc^2)^2$; from this, we have the energy term mc^2 itself being invariant; it is an energy that is the same in all frames of reference.

If we interpret this not just as mathematical but also as having physical meaning, then this would be a new type of energy associated with every particle, previously unknown before special relativity. This energy does not depend on the speed of a particle; it is an intrinsic energy all masses have, even when at rest. We call this the rest mass energy, written as

$$E_0 = mc^2 \quad (3.20)$$

From this, we can deduce that if the mass of a system of particles decreases, then the decrease in energy due to the loss of mass must be offset by an increase in other forms of energy to keep the system's energy conserved. For example, this can be done by converting the energy previously stored in its mass into energy in the form of light. This conversion of energy is consistent with experimental results.

Although this equivalence has been misused to make nuclear bombs, you can thank it for the ability to generate energy from decaying radioactive elements in nuclear power plants and for the future possibility of generating energy from nuclear fusion plants. It also helps explain why stars can produce so much energy, as they convert some of their mass into energy in the form of

light, through nuclear fusion reactions.

Now, looking at the rest mass energy equation, it does not seem like a very proper way to derive this famous equation, or at least it looks like we have taken a shortcut. But let us continue, and we will look at what the other terms in equation (3.19) represent, to see if we get anything else that may convince us a little more that this is an okay interpretation of this invariant quantity of energy.

To do this, we will first derive the relativistic version of momentum.

3.3.3 Relativistic Momentum

We will now look at the other two terms in equation (3.19); both contain the gamma function, and we will use the Taylor series approximation to obtain more information about each term.

derivation:

So let us rearrange equation (3.19), to get

$$(\gamma_u mc^2)^2 = (mc^2)^2 + (\gamma_u mu_q c)^2 \quad (3.21)$$

for the energy term in the brackets on the left-hand side, we can use what is called the Taylor expansion to write γ_u as a convergent series:

$$\gamma_u \equiv 1 + \frac{1}{2} \frac{u_q^2}{c^2} + \frac{3}{8} \frac{u_q^4}{c^4} + \dots \quad (3.22)$$

Giving the energy term on the left-hand side as

$$\begin{aligned} \gamma_u mc^2 &= mc^2 \left(1 + \frac{1}{2} \frac{u_q^2}{c^2} + \frac{3}{8} \frac{u_q^4}{c^4} + \dots \right) \\ &= mc^2 + \frac{1}{2} mu_q^2 + \frac{3}{8} m \frac{u_q^4}{c^2} + \dots \end{aligned} \quad (3.23)$$

The first quantity in this equation is an energy that does not depend on the particle's speed. It is the invariant energy from before, the rest mass

energy. The second quantity is the classical kinetic energy formula. The rest may be viewed as higher-order kinetic energy terms, which are absent in classical physics because they are very small for $u_q \ll c$. So this seems like it could be a formula for the particle's total energy (in a vacuum, with no potential energy). We will denote this total energy as

$$E = \gamma_u mc^2 \quad (3.24)$$

and the rest mass energy from the last section, denoted as E_0 , allowing us to write equation (3.21) as

$$E^2 = E_0^2 + (\gamma_u mu_q c)^2 \quad (3.25)$$

Now, let us look at this last energy term in the brackets, along with the Taylor expansion of γ_u , giving

$$\begin{aligned} \gamma_u cmu_q &= cmu_q \left(1 + \frac{1}{2} \frac{u_q^2}{c^2} + \frac{3}{8} \frac{u_q^4}{c^4} + \dots \right) \\ &= c \left(mu_q + \frac{1}{2} m \frac{u_q^3}{c^2} + \frac{3}{8} m \frac{u_q^5}{c^4} + \dots \right) \end{aligned} \quad (3.26)$$

At small particle velocities, this last term is approximately $(mu_q)c$ with mu_q being the classical momentum; it seems like this energy could be the relativistic version of the particle's momentum, times the speed of light.

So we will think of the last term in equation (3.25) as the relativistic momentum of the particle:

$$p = \gamma_u mu_q \quad (3.27)$$

which is approximately the classical momentum at speeds much less than the speed of light.

3.3.4 Proof that the relativistic Momentum is Conserved

Classically, we have conservation laws for momentum and energy, meaning that for an isolated system, both remain constant. For an isolated system of particles, the total energy and the momentum of all the particles before an interaction is the same as the total energy and the momentum of all the particles after,

$$\begin{aligned}p_{after} &= p_{before} \\ E_{after} &= E_{before}\end{aligned}\tag{3.28}$$

In classical physics, we only cared about these two quantities because they have these conservation laws; if they did not, they would not be useful. Hence, we would like for there to still be conservation laws in special relativity for the relativistic versions of the quantities. And for these conservation laws to hold, we would need them to hold in all inertial frames of reference. So we will show that if the total energy and relativistic momentum are conserved in one frame, then they are conserved in all reference frames.

derivation:

We will start with the previous definitions of momentum and total energy: $\underline{p} = \gamma_u m \underline{u}$ and $E = \gamma_u mc^2$.

Transformation Identities

We choose the primed frame to move in the z-direction as usual, giving the velocity transform between the frames from equation (2.44). We will also make use of an identity for $\gamma_{u'}$ given by

$$\gamma_{u'} = \gamma_u \gamma \left(1 - \frac{u_z v}{c^2}\right)\tag{3.29}$$

I will let you check this identity yourself, as it is just the substitution of the transform equation of u' into the gamma function, with some

rearrangement.

Y-Component

Using equation (2.44) and equation (3.29) on the relativistic momentum's y-component in the primed frame $p'_y = \gamma_{u'} m u'_y$, gives:

$$\begin{aligned} p'_y &= \left[\gamma_u \gamma \left(1 - \frac{u_z v}{c^2} \right) \right] m \left[\frac{u_y}{\gamma \left(1 - \frac{u_z v}{c^2} \right)} \right] \\ p'_y &= \gamma_u m u_y \\ p'_y &= p_y \end{aligned} \tag{3.30}$$

In the y-direction, the equation for the momentum is unchanged. Therefore, if the momentum is conserved before and after an event in the initial frame, it is still conserved in the primed frame. This new relativistic momentum is required to allow for the conservation law to work between different frames for the y-component.

Z-Component

Now doing the same for the z-component $p'_z = \gamma_{u'} m u'_z$, we get:

$$\begin{aligned} p'_z &= \left[\gamma_u \gamma \left(1 - \frac{u_z v}{c^2} \right) \right] m \left[\frac{u_z - v}{1 - \frac{u_z v}{c^2}} \right] \\ p'_z &= \gamma_u \gamma m (u_z - v) \\ p'_z &= \gamma (\gamma_u m u_z) - \gamma v (\gamma_u m) \end{aligned} \tag{3.31}$$

Proof of Conservation

Using the relativistic momentum in the initial frame $p_z = \gamma_u m u_z$ and the equation for the total energy that we described in the previous section $E = \gamma_u m c^2$, we obtain:

$$p'_z = \gamma \left(p_z - \frac{v E}{c^2} \right) \tag{3.32}$$

now, if we do have the total momentum and total energy before and after conserved, we have

$$\begin{aligned}\Delta p'_z &= \gamma \left(p_{z\text{after}} - \frac{v E_{\text{after}}}{c^2} \right) - \gamma \left(p_{z\text{before}} - \frac{v E_{\text{before}}}{c^2} \right) \\ &= \gamma \left(\Delta p_z - \frac{v \Delta E}{c^2} \right)\end{aligned}\tag{3.33}$$

This last equation is the transformation of the momentum in the z-direction of each particle to a primed frame.

In the isolated system in the initial frame, we want both the momentum and energy to be conserved

$$\begin{aligned}\Delta p_z &= 0 \\ \Delta E &= 0\end{aligned}\tag{3.34}$$

If these are indeed conserved, then

$$\Delta p'_z = 0\tag{3.35}$$

giving the momentum conserved in the z-direction in the primed frame as well.

So the momentum is conserved in all the components in the primed frame. Therefore, taking the $\gamma_u m u_q$ term as being the relativistic momentum and the term $\gamma_u m c^2$ as the total energy is consistent with special relativity and the laws of the conservation of momentum and energy.

Now, what about for light itself, when γ is infinitely large, can we still get the energy-momentum relation to work?

3.3.5 Momentum of Light

Here, we will show how to get the only reasonable momentum equation of light that we can from the energy-momentum relationship equation.

So let us start by rewriting the energy-momentum equation (3.25).

$$E^2 = (mc^2)^2 + (\gamma_u mu_q c)^2 \quad (3.36)$$

For the speed of light, we have γ_u being infinitely large, so if the photon had a non-zero rest mass, the second term would be infinite, therefore giving the total energy as infinite, which is not possible. So the photon must have zero (or at least only infinitesimal) rest mass, giving zero for the first term (rest mass energy), reducing the total energy of a photon to;

$$E_{ph} = (\gamma_c m_{ph} c) c \quad (3.37)$$

now, if we were to get a finite energy from this, we would require $\gamma_c m_{ph}$ (which is an infinite value multiplied by an infinitesimal) to be finite for it to give a meaningful energy of the photon. We previously used the term $p = \gamma_u m$ to represent the relativistic momentum of a particle. So we could take this term as representing the photon's momentum as well, and think of $m_{eff} = \gamma_c m_{ph}$ as the "effective mass" that a photon has, giving its momentum as $p_{ph} = m_{eff} c$. Therefore, the momentum for light in terms of its energy as

$$p_{ph} = \frac{E_{ph}}{c} \quad (3.38)$$

with p_{ph} and E_{ph} being the momentum and energy of the photon.

This equation for the energy-momentum relationship of light can also be derived in electromagnetic theory [5], which gives us more confidence that we are correct.

The last equation allows us to generalize the last energy term in equation (3.25), as the momentum times the speed of light

$$E_p = pc \quad (3.39)$$

giving us the final piece of the puzzle.

3.3.6 Energy-Mass-Momentum Relation

Now that we have confirmed the use of relativistic momentum from equation (3.27), we can substitute it and equation (3.20) into (3.25) to finally obtain the famous energy-momentum relationship:

$$E^2 = (mc^2)^2 + (pc)^2 \quad (3.40)$$

with momentum being $p = \gamma_u m u_q$ for a particle with mass, and written more meaningfully for a photon as $p = E/c$.

Side Note:

When looking at particle collisions, we also need to take into account potential energy as particles move towards and away from each other, but this must take into account for acceleration as well. which is not an easy thing to do.

3.3.7 Speed of Light Limit

Previously, we showed that faster than light travel would lead to effects being able to occur before their cause, in section (2.8), but here is another reason why faster than light transport may not be possible for particles with mass. We can see from the total energy equation (3.24), that if a particle with mass were accelerated to the speed of light, its energy would become infinite. So we would need infinite energy (which does not exist) to get that particle to the speed of light. Therefore, it is not possible.

If you still want to consider the possibility of a particle moving at speeds faster than light speed, it would lead to γ being imaginary (the square root of a negative number). But we require that energy be a real quantity for it to be measurable, and for this, we would need the mass to be imaginary as well. Some Physicists theorized a particle with an imaginary mass, the tachyon, but this still has the problem that cause and effect are interchangeable, as we explained in section (2.8). The tachyon has never been detected in any detectors.

3.4 Summary

In this chapter, we looked at the space-time interval, which is a very important quantity in special relativity. It is the simplest quantity that is invariant when switching between inertial reference frames (It has the same value in all inertial frames of reference). From this, we derived the D'Alembert operator, an important operator in relativistic wave equations. We showed that it is also invariant when switching between inertial reference frames. The space-time interval also allowed us to find an invariant quantity with the units of energy. This led us to the concepts of a particle's rest mass energy, the total relativistic energy, and relativistic momentum. From this, we showed that these were consistent with the conservation laws for energy and momentum; if they were true in one inertial frame, they then hold in all inertial frames. Next, we found the relation between energy and momentum for light, which finally enabled us to write the energy-momentum equation in a more meaningful way.

In this chapter, we learned that mass and energy are equivalent, that particles with mass would require infinite energy to be accelerated to the speed of light, and that our definition of momentum needed to be updated to apply to light as well.

What we have learned here brings us to the boundary between pure special relativity and particle physics, which relies heavily on experimental physics to discover the rules that particles follow. Interactions between particles that produce other particles, require more than just conservation of energy and momentum to fully describe them; there is also the conservation of charge, lepton number, baryon number, etc., which determine which interactions can and can not happen. Hence, this is outside the context of this book.

So we will move on to Part (III), where we will take a close look at how light's propagation behaves in special relativity. We will first examine the delayed view of the world an observer sees due to the delay in light signals from their surroundings, and then move on to how light is affected when emitted from a

moving point source, leading to many insights, including the famous Doppler effect.

Part III

Light in Special Relativity

Here, we will discuss the propagation of light and its properties. We will look at the delay in light's signal due to the finite speed of light, and the effects of the principles of special relativity on the emission of light from a moving source. A lot of the concepts in the Chapters of this Part are not taught in most resources that you will find on special relativity, even though they can be very insightful. The exception to this is the Doppler effect on light's frequency, which is discussed in most resources and is derived here.

Chapter 4

An Observer's Delayed Worldview

Light travels extremely fast compared to other everyday speeds we are used to, roughly at 300,000,000 meters per second. In the blink of an eye, light could travel around the world twice. So we do not normally need to worry about delays in the light signal, as the time for light to propagate over the distances in our everyday world is extremely short. Therefore, the signal can be approximated as being instantaneous, and we can ignore the more complex modeling of the delayed light signal. But when it comes to particle physics or astronomy, we might be dealing with speeds close to that of light or huge distances. For example, light has taken 152,000 years to travel from the Andromeda Galaxy, so we are currently seeing the Andromeda Galaxy where it was and how it looked 152,000 years ago. Their past positions, which we currently see them in, are called their retarded positions. The topic in this chapter is not normally included in Special relativity texts. Still, it is necessary to give you a fuller story and better understanding of relativity. These topics provide additional insights into why the coordinate transform equations have to be the way they are. As we will see when transforming

retarded positions between frames.

4.1 Retarded Coordinates

In an initial frame, we have its own coordinates at rest. If a light signal is emitted from a coordinate to an observer at the origin, it takes time to reach the observer, so the observer would see the stationary coordinate at the previous time that the light was emitted from this coordinate. This previous time is the position coordinate's retarded time.

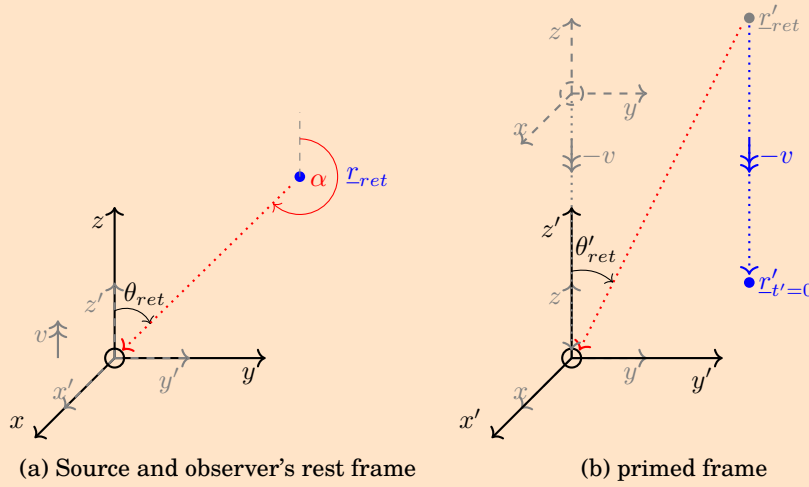


Figure 4.1: Retarded coordinates. Diagrams of a source of light that represents, in the source's rest frame, the coordinate $\underline{r} = \underline{r}_{ret}$ (blue) emitting light (red) at an angle α towards the origin, which is received by an observer positioned at the origin at time $t = t' = 0$ with the observer and source at rest in the first inertial frame (left) and then shown moving in a primed frame at speed $-v$ in the z -direction (right). In the primed frame, the past, retarded position of the rest frame and also the source's coordinate $\underline{r}'_{ret}$ at its retarded time are shown in gray, with the source's current position $\underline{r}'_{t'=0}$ at time $t' = 0$ shown in blue. The path light takes is shown in red. The corresponding axis in each frame is also shown in black. The angle of the retarded position from the origin is θ_{ret} and θ'_{ret} , respectively.

As shown in the diagram, if we take the rest frame of an observer and a source that emits a pulse of light from its position at $\underline{r} = \underline{r}_{ret}$ so that it is

received by the observer at the origin, at time $t = 0$. The retarded time the pulse of light was emitted is

$$t_{ret} = -\frac{\|\underline{r}_{ret}\|}{c}, \quad (4.1)$$

which is the negative of the time it took for light to propagate along $\underline{r}_{ret}$ to the origin. The velocity of the light that propagated to the origin is

$$\underline{c} = c \begin{pmatrix} 0 \\ \sin \alpha \\ \cos \alpha \end{pmatrix} \quad (4.2)$$

Where α is the direction of propagation. This velocity and the retarded time (which is negative) give the retarded position at which the light was emitted:

$$\underline{r}_{ret} = -\|\underline{r}_{ret}\|\hat{\underline{c}} = ct_{ret} \begin{pmatrix} 0 \\ \sin \alpha \\ \cos \alpha \end{pmatrix} \quad (4.3)$$

The source's current position and its retarded position are the same in its rest frame, since the source is not moving. Also, the position vector and the light's velocity vector are in opposite directions. A negative emission time reverses the direction of the velocity vector here. Now, inputting this retarded position and time into the coordinate transform, we get the retarded position and time in the primed frame as

$$\underline{r}'_{ret} = ct_{ret} \begin{pmatrix} 0 \\ \sin \alpha \\ \gamma \left(\cos \alpha - \frac{v}{c} \right) \end{pmatrix} = \begin{pmatrix} 0 \\ y \\ \gamma (z - vt_{ret}) \end{pmatrix} \quad (4.4)$$

at retarded time

$$t'_{ret} = \gamma \left(1 - \frac{v}{c} \cos \alpha \right) t_{ret}$$

If instead we use the angle of the retarded coordinate from the z-axis, $\theta_{ret} =$

$\alpha - \pi$, we have that $\cos \alpha = \cos(\theta_{ret} + \pi) = -\cos \theta_{ret}$ and therefore the retarded time can be given as

$$t'_{ret} = \gamma \left(1 + \frac{v}{c} \cos \theta_{ret} \right) t_{ret} \quad (4.5)$$

With the light being received at the origin of the primed frame axis at time $t' = 0$.

The coordinate transform here for retarded coordinates is the normal coordinate transform, except the time transform simplifies and allows us to take the retarded time out of brackets with just the factor $\mathring{A} = \gamma \left(1 - \frac{v}{c} \cos \alpha \right) = \gamma \left(1 + \frac{v}{c} \cos \theta_{ret} \right)$ now in front of it. This factor will also come up in the Doppler effect, in aberration, and when finding the density of light from a source that is emitting it continuously in all directions. We can use this factor to simplify equations as we did with γ previously. We will call $\mathring{A}$ the aberrational factor, also known as the relativistic Doppler factor.

When switching to the primed frame, the angles at which light is emitted and received change; this is called the relativistic aberration of light. This aberration will be described fully in the next chapter.

Side Note:

This retarded position of the source is at its retarded time. If you wanted to find the position of the source at the same time as the observer, $t' = 0$, we need to propagate it from its retarded position and time to this time of zero. So we need to take into account that the source is moving at $u = -v$ and the retarded time is the negative of the time that we want to pass, for the source to move to its current position. These negatives cancel out. Which gives the current position, with a bit of working out, as

$$\underline{r}'_{t'=0} = \underline{r}'_{ret} + \underline{v}t'_{ret} = ct_{ret} \begin{pmatrix} 0 \\ \sin \alpha \\ \frac{1}{\gamma} \cos \alpha \end{pmatrix} \quad (4.6)$$

So for a stationary source in the rest frame at $t = 0$, we have this position in

the primed frame at $t' = 0$.

4.2 The Transformed Retarded Coordinates

If we treat the coordinates of a coordinate system as retarded sources of light in the same way as we did for the source in the previous section, we can transform these retarded coordinates of the initial frame into the primed frame. This transform is shown in the figure below for a small subset of coordinates, to give you an idea of how the initial frame's retarded coordinates are spread in the primed frame.

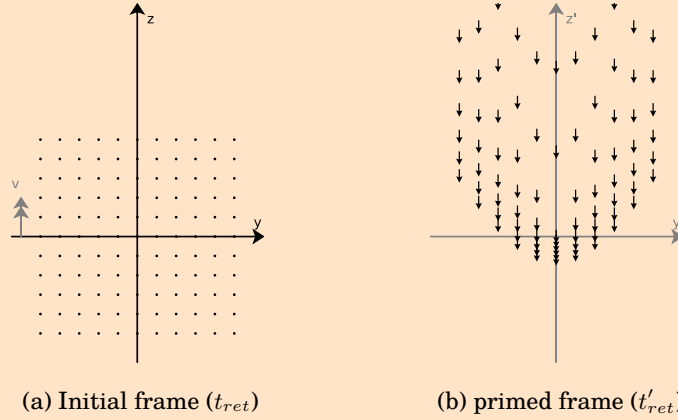


Figure 4.2: Retarded coordinate transform. A diagram, showing an initial small grid of points, representing some position coordinates that are at rest in their proper frame (left), with an observer at the origin, using the retarded time $t_{ret} = \frac{\|r\|}{c}$ for these coordinates, we can transform these to their retarded positions in a primed frame using the coordinate transform, with the original frames coordinates now moving at velocity $-v = -0.9$, the initial frames coordinates are transformed to where they would be seen to be at by an observer at the origin with the observers time $t' = 0$, these positions are due to the delay in the light from these points taking time to get to the origin, they are also known as the retarded positions.

The density of the initial frame's retarded coordinates in its own frame is constant everywhere, but in the primed frame, the initial frame's coordinates have a density that depends on the angle, as shown in the figure below. This

figure is shown but not derived, as this requires mathematics from later in section (6.2).

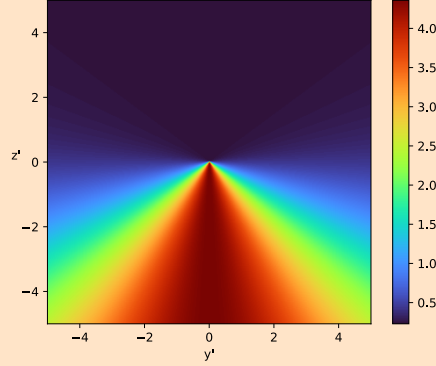


Figure 4.3: **Volume density of initial frame's retarded coordinates in primed frame.** This diagram shows the density of the initial frame's coordinates at their retarded positions in the primed frame, with $v = 0.9$, using a color scale.

If we are at rest in the initial frame, and we see the points as they are shown in the initial frame in figure (4.2a), then if there were another observer overlapping with us moving at $\underline{v}$, they would see all the points distributed as they are shown in their primed frame in figure (4.2b).

4.3 Apparent Velocity

From the previous section, we saw that objects appear to be at their retarded positions due to the delay in the light signal from them. That would also mean that we have the rate of change of the observed positions differing from their actual rate of change. That is, the apparent velocity of an object is different from its actual velocity. This concept of apparent velocity was demonstrated in section (1.13), where we saw that objects can be perceived as moving faster than light even if they are not actually doing so. Here, we will show how to find the Apparent velocity of objects.

If we have a moving particle at its retarded position $\underline{r}_{ret}$, then in an infinites-

imal time dt it will move by a displacement of $\underline{u}_{ret}dt$ to position $\underline{r}_{ret} + \underline{u}_{ret}dt$, as shown in the Diagram below (where $\underline{u}_{ret}$ is the velocity of the particle at the retarded time).

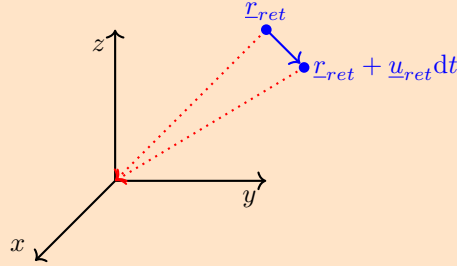


Figure 4.4: **Apparent velocity.** Diagram showing a particle (blue) moving from a retarded position $\underline{r}_{ret}$ by a displacement of $\underline{u}_{ret}dt$ (with $\underline{u}_{ret}$ being the velocity of the particle at the retarded time, and dt being an infinitesimal change in time), the paths that light has to travel to get to the origin from both these positions are shown (red). Leading to a different retarded time between the positions than the actual time between them. This is because the particle moves position, requiring the light signal to travel a different distance to the observer, which introduces an additional time difference. This leads to a different Apparent velocity to the actual velocity.

The difference between the retarded times of light from each position reaching the origin is the time that light takes to get from position $\underline{r}_{ret} + \underline{u}_{ret}dt$ minus the time it takes to get from position $\underline{r}_{ret}$, plus the time it takes for the particle to move to the new position, dt . Which is

$$dt_{ret} = \frac{\|\underline{r} + \underline{u}_{ret}dt\|}{c} - \frac{\|\underline{r}\|}{c} + dt \quad (4.7)$$

Since $\underline{u}_{ret}dt$ is infinitesimally small, we can simplify this expression by using some approximations as shown in the following derivation.

$$\begin{aligned}
\|\underline{r}_{ret} + \underline{u}_{ret}dt\| &= \sqrt{(\underline{r}_{ret} + \underline{u}_{ret}dt) \cdot (\underline{r}_{ret} + \underline{u}_{ret}dt)} \\
&= \sqrt{\underline{r}_{ret} \cdot \underline{r}_{ret} + 2\underline{r}_{ret} \cdot \underline{u}_{ret}dt + \underline{u}_{ret}^2 dt^2} \\
&\approx \sqrt{\underline{r}_{ret} \cdot \underline{r}_{ret} + 2\underline{r}_{ret} \cdot \underline{u}_{ret}dt} \\
&\approx \|\underline{r}_{ret}\| \sqrt{1 + \frac{2\underline{r}_{ret} \cdot \underline{u}_{ret}dt}{\underline{r}_{ret} \cdot \underline{r}_{ret}}} \\
&\text{using the Taylor expansion } \sqrt{1+N} \approx 1 + \frac{N}{2} \text{ for } N \ll 1: \\
&\approx \|\underline{r}_{ret}\| \left(1 + \frac{\underline{r}_{ret} \cdot \underline{u}_{ret}dt}{\|\underline{r}_{ret}\|^2} \right) \\
&\approx \|\underline{r}_{ret}\| + \frac{\underline{r}_{ret} \cdot \underline{u}_{ret}dt}{\|\underline{r}_{ret}\|} \\
&\approx \|\underline{r}_{ret}\| + \hat{r}_{ret} \cdot \underline{u}_{ret}dt
\end{aligned} \tag{4.8}$$

So, for the infinitesimal change in retarded positions we have $\|\underline{r}_{ret} + \underline{u}_{ret}dt\| = \|\underline{r}_{ret}\| + \hat{r}_{ret} \cdot \underline{u}_{ret}dt$, which we can substitute into equation (4.7) to get the time that has passed between the two positions as observed from the origin i.e., the perceived retarded time that has passed at the origin, given as

$$\begin{aligned}
dt_{ret} &= dt + \frac{\hat{r}_{ret} \cdot \underline{u}_{ret}dt}{c} \\
&= dt + \frac{\hat{r}_{ret} \cdot \underline{u}_{ret}dt}{c} \\
&= \left(1 + \frac{\underline{u}_{ret} \cdot \hat{r}_{ret}}{c} \right) dt
\end{aligned} \tag{4.9}$$

The distances between the points are the same for the observer as the true distances between them; it is just the perceived retarded times that differ. Therefore, we can get this apparent velocity as

$$\begin{aligned}
\underline{u}_{app} &= \frac{\underline{u}_{ret} dt}{dt_{ret}} \\
&= \frac{\underline{u}_{ret} dt}{\left(1 + \frac{\underline{u}_{ret} \cdot \hat{r}_{ret}}{c}\right) dt} \\
&= \frac{\underline{u}_{ret}}{1 + \frac{\underline{u}_{ret} \cdot \hat{r}_{ret}}{c}}
\end{aligned} \tag{4.10}$$

This equation shows that the apparent velocity can be much larger or smaller, depending on the direction of motion of the object. For something moving at the speed of light, we have the apparent velocity being between $0.5c \leq \|\underline{u}_{app}\| \leq \infty$ depending on the direction it is moving.

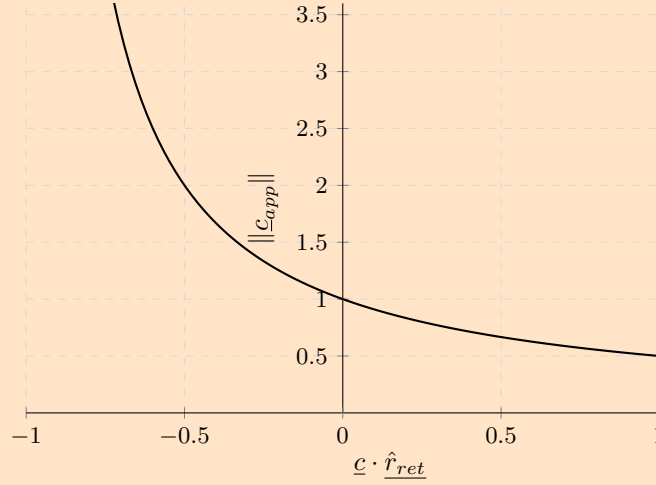


Figure 4.5: **Apparent speed of light.** A diagram, showing the magnitude of the apparent velocity of light $\|\underline{c}_{app}\|$ depending on the direction of light relative to the radial direction from the observer, that is $-1 \leq \underline{c} \cdot \hat{r}_{ret} \leq 1$.

This section highlights an important concept that needs to be taken into account when making some experimental observations, such as the motion of stars, planets, and galaxies in astrophysics.

4.4 Summary

In this chapter, we introduced retarded positions, which are the positions an observer perceives objects to currently be at due to the delay in the light signal from the object to the observer, with their associated retarded time, which is the negative of the propagation time light takes to travel from the retarded position to the observer. We also derived how to transform these retarded coordinates to the corresponding prime frame's retarded coordinates. The retarded coordinate transform and also the density of the transformed initial frame's retarded coordinates in the primed frame were shown in figure (4.2) and figure (4.3) respectively. We then derived the apparent velocity of objects, which is the velocity an observer perceives an object to be moving at, which is consequence of the delay in the light signal. This velocity differs from the object's actual speed.

What we have learned in this chapter will be used later when performing transformations on light pulses.

Chapter 5

A Point Source: Emitting a Pulse of Light

If a light pulse is emitted from a point source in all directions in its rest frame, how will this be affected in a frame where the source is now moving? Answering this question will be the topic of this chapter. For this, we will look at the velocity transform specifically for the speed of light, which will be used throughout this chapter, then move on to look at the effect the transform has on the frequency of light that a source emits, the "Doppler effect". Next, we will get to "relativistic aberration", which explains how the angle at which the emitted light is propagated is affected. The combined effects of these together is called "relativistic beaming". Knowing all this will allow us to transform a field of continuously emitted light from a source in the next chapter and determine its properties. But to allow us to do all this, we will first need to describe the mathematics of spherical polar coordinates. The topics of relativistic aberration and beaming in this chapter are not covered in most special relativity sources. But they are interesting and offer important insights. So, They can be worth your while to learn.

5.1 Spherical Polar Coordinates

Before we go any further with special relativity, we will need to take some time making sure we know the math needed for what comes next. This section will be about spherical polar coordinates, which will help us describe how the propagation of light changes between frames, by allowing us to explain aberration, which is how the propagation angle changes between frames, and the Doppler effect, which is how the wavelength of light changes between frames. So let us start with how to describe a vector in terms of the spherical polar coordinates, which is shown in the following figure in red:

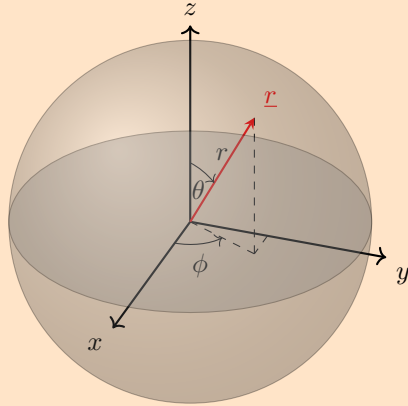


Figure 5.1: **Spherical polar coordinates.** Diagram of the Cartesian x, y, z -axis showing a position vector $\underline{r}$ (red) with its associated spherical polar coordinates (r, θ, ϕ) , where θ is the angle from the z -axis, ϕ is the angle from the x -axis in the x - y plane, and r is the magnitude.

The figure shows that, for a given position $\underline{r}$, we can write it in Cartesian coordinates as (x, y, z) , or in spherical polar coordinates as (r, θ, ϕ) . Where the angles from the Z -Axis and X -axis are θ (between 0 and π) and ϕ (between 0 and 2π) respectively, and r is the displacement to the position from the origin.

The position can be written using either coordinate system as follows

$$\underline{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = r \begin{pmatrix} \cos \phi \sin \theta \\ \sin \phi \sin \theta \\ \cos \theta \end{pmatrix} \quad (5.1)$$

with

$$\begin{aligned} r = \|\underline{r}\| &= \sqrt{x^2 + y^2 + z^2} \\ \cos \theta &= \frac{z}{r} \\ \cos \phi &= \frac{x}{\sqrt{x^2 + y^2}} \end{aligned} \quad (5.2)$$

This is all we shall need to know about spherical polar coordinates for this chapter. We will need to derive some more math beyond this in the next chapter, but we will look at it when it comes up.

5.2 Velocity Transform of Light from a Source

From here, we will explicitly use the source's rest frame and the frame in which the source is moving at speed u . We will denote the terms in the rest frame with a subscript zero (i.e., term₀), and the moving frame with unaltered notation, as it will be the moving frame that we will be interested in finding information about, and it will allow to read equations for this chapter and the next more easily. Think of it as the frame with the moving source being the current frame and the rest frame being the one we need to compare to. Now, let us get back to it.

The speed of light remains the same after transforming between frames, but the velocity does not, because the direction of propagation changes. Light emitted at a speed c from a source at rest, at an angle α_0 from the z-axis and φ_0 from the x-axis, has the velocity written in spherical polar coordinates as

$$\underline{c}_0 = c \cdot \hat{\underline{c}}_0 = c \begin{pmatrix} \cos \varphi_0 \sin \alpha_0 \\ \sin \varphi_0 \sin \alpha_0 \\ \cos \alpha_0 \end{pmatrix}, \quad (5.3)$$

Then, for the frame in which the source moves at speed u in the z -direction, we have this frame moving at $v = -u$ relative to the rest frame. So, using the velocity transform from equation (2.44), we get the light's velocity in this frame as

$$\underline{c} = c \begin{pmatrix} \cos \varphi \sin \alpha \\ \sin \varphi \sin \alpha \\ \cos \alpha \end{pmatrix} = \frac{c}{\gamma \left(1 + \frac{u}{c} \cos \alpha_0\right)} \begin{pmatrix} \cos \varphi_0 \sin \alpha_0 \\ \sin \varphi_0 \sin \alpha_0 \\ \gamma \left(\cos \alpha_0 + \frac{u}{c}\right) \end{pmatrix} \quad (5.4)$$

Using a bit of trigonometry to find the z -components in terms of sines instead of cosines, and substituting them into the other two components, you can find that $\varphi = \varphi_0$.

This frame's velocity still has the speed of light, which can be seen by taking the magnitude of the velocity vector. However, the angle of propagation from the z -axis has changed; the change in this angle is described from the z -components of the equation above, as

$$\cos \alpha = \frac{\left(\cos \alpha_0 + \frac{u}{c}\right)}{\left(1 + \frac{u}{c} \cos \alpha_0\right)} \quad (5.5)$$

This is the relativistic aberration formula [2], which shows how the direction of light's propagation transforms, as shown in Figure (5.2). Rearranging this equation gives us the coordinate conversion in the opposite direction.

$$\cos \alpha_0 = \frac{\left(\cos \alpha - \frac{u}{c}\right)}{\left(1 - \frac{u}{c} \cos \alpha\right)} \quad (5.6)$$

These equations, which show how the cosine of the propagation angle of the emitted light changes, will be put to use in the next section.

5.3 Aberration in Special Relativity

We have just derived how the propagation angle of light changes when switching frames. Here, we shall visually demonstrate the changes in propagation angle by looking at the following situation.

If a source in its rest frame emits light evenly in all directions, then each emitted light's angle of propagation is transformed using equation (5.4). The angles between the light's propagation and the z-axis, in the source's moving frame, are found using equation (5.5). This is shown in the following diagram.

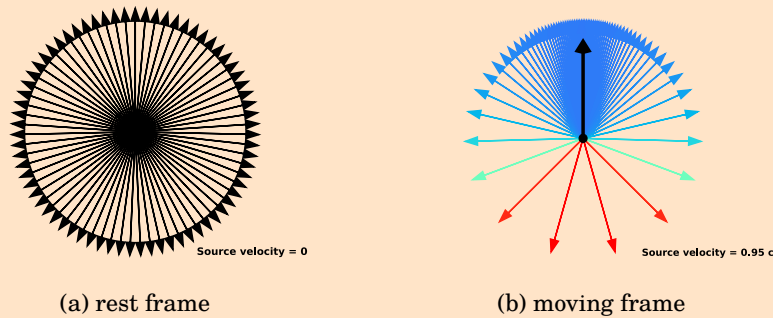


Figure 5.2: **Emitted light in two frames.** Diagram showing the directions of propagation of an evenly distributed light pulse in the rest frame of a source (left) and the corresponding distribution of the light pulse's propagation directions in a moving frame of the source (right), with the colors showing the Doppler effect on the light's frequency, which will be explained in the next section.

The diagram shows that the moving source emits more radiation in the direction of its motion, compared to its rest frame. This is called relativistic aberration. If we want to see the opposite, with light instead being absorbed instead of emitted, we can look at the following figure.

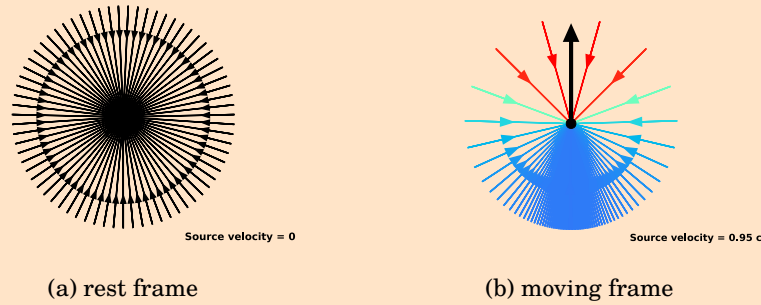


Figure 5.3: **Absorbed light in two frames.** Diagram showing the directions of evenly distributed light being absorbed by a receiver in its rest frame (left) and the corresponding distribution of the light pulse's propagation directions in a frame in which the receiver is now moving (right), with the colors showing the Doppler effect on the light's frequency, which will be explained in the next section.

For the case of a source, as its speed tends to the speed of light, u/c tends to 1, and from equation (5.5), we can see that, as a result of this, $\cos \alpha$ tends to 1. This means that all emitted light from the source tends to the direction of its motion as its speed tends to c . We also have that the flow of time for a moving source relative to its rest frame will become infinitely slow, coming to a stop, when moving at the speed of light. That is, the act of emitting the light stalls. For both these reasons, the light would remain in the vicinity of the source. I will leave this for you to contemplate.

5.4 Doppler Effect

When we switch between frames, the wavelength of light changes. This is called the Doppler effect and was shown by the color shift in the figures from the previous section. We will describe this effect using the help of the following simple figure of light being emitted from a source in its rest frame and in a frame in which it is moving.



Figure 5.4: **The Doppler effect.** Diagrams showing a source emitting light at a velocity c_0 in the source's rest frame (left) and a second frame where the source is now moving at velocity u with the light now being emitted at an angle α , giving the light's velocity as c . The different colors of light in each frame show how the Doppler effect changes the light's frequency.

In the rest frame of the source, if it emits a wave at the speed of light in any direction, we have the length the wave has traveled from the source in the infinitesimal time dt_0 as being

$$d\lambda_0 = c dt_0 \quad (5.7)$$

If we change to a frame where the source is moving at velocity u in the z -direction, the wave is emitted at an angle α from the z -axis. It travels a distance $c dt$ in a time dt , with the source moving a distance $u dt \cos \alpha$ in the direction of the emitted wave. It leads to a bunching up of the wave in the direction of the movement of the source, as the same amount of the wave has been emitted in this frame's corresponding time, but the distance it has traveled from the source is

$$\begin{aligned} d\lambda &= c dt - u \cos \alpha dt \\ &= \left(1 - \frac{u}{c} \cos \alpha\right) c dt \\ &= \gamma \left(1 - \frac{u}{c} \cos \alpha\right) c dt_0 \end{aligned} \quad (5.8)$$

With $dt = \gamma dt_0$ being used in the last step, from the time dilation equation (2.15) (using $\Delta t_0 = dt_0$ and $\Delta t = dt$ substituted in. The ratio of the displacement of the wave compared to its displacement in the rest frame is

$$\frac{d\lambda}{d\lambda_0} = \gamma \left(1 - \frac{u}{c} \cos \alpha \right) \quad (5.9)$$

Wavelength is inversely proportional to the frequency, and this bunching up at the front will proportionally decrease the wavelength. The frequency ratio ν/ν_0 is then

$$\frac{\nu}{\nu_0} = \frac{d\lambda_0}{d\lambda} = \frac{1}{\gamma \left(1 - \frac{u}{c} \cos \alpha \right)} \quad (5.10)$$

This gives us the formula for the frequency of light emitted by a moving source relative to its frequency in the source's rest frame.

This frequency shift can be used in astronomy to determine how fast stars and galaxies are moving relative to Earth. It is done by measuring the frequency of light from the star/galaxy and comparing it to the expected frequency if it were at rest, with the expected frequency calculated using some astrophysics, which we won't go into here. But for those who are interested, there are two videos explaining how we work such things out in astrophysics (*part 1, part 2*).

5.4.1 Frequency Change from switching Frame

The previous section shows how to calculate the frequency of light in a source's moving frame relative to its rest frame. But, if you want the frequency change of light, when you switch from any initial frame to a primed frame, then you can get it in terms of the primed frame's velocity v as

$$\frac{\nu'}{\nu} = \gamma \left(1 - \frac{v}{c} \cos \alpha \right) = \frac{1}{\gamma \left(1 + \frac{v}{c} \cos \alpha' \right)} \quad (5.11)$$

5.5 Relativistic Beaming

Relativistic beaming is the combination of both effects from the two previous sections. To show it, we will look at a light pulse emitted from a source in

its rest frame and how it appears in a frame where the source is moving. First, by applying the Lorentz transform to the pulse and source, and then propagating the pulse backwards or forwards in time to sync the light's times with the source at the origin.

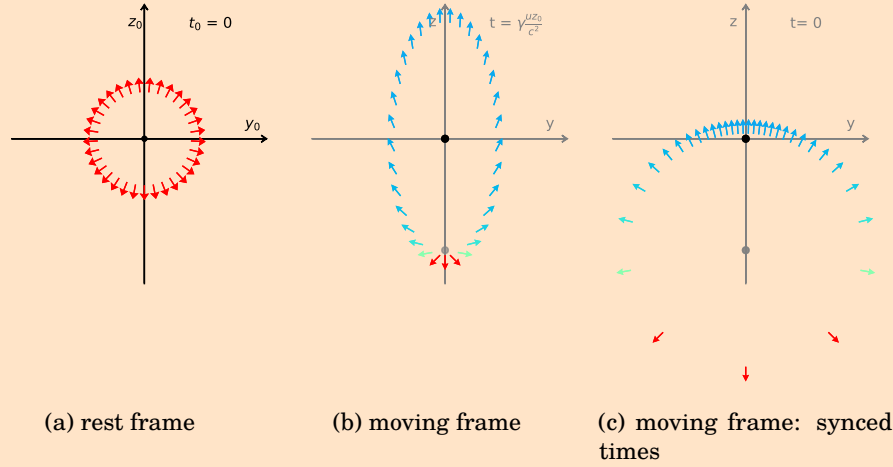


Figure 5.5: Relativistic beaming. Diagram showing, the position of an evenly spherically distributed light pulse in the source's rest frame (left) after propagating for a certain amount of time, with its current time $t_0 = 0$, it is also shown in the moving frame using the Lorentz coordinate transform (middle) giving an ellipsoid with unsynchronized times, and also shown in the moving frame with each part of light pulse propagated forward or backwards in time to the origins time $t = 0$ (right). The black dot shows the current location of the source, while the gray dot shows its position when the pulse was emitted. All the propagation of the light is in the direction that is outward from this point.

The pulse's positions in the moving frame, with pulse's times synced with the source at the origin are shown in figure (5.5c). They can be found from the Lorentz transformed rest frame's coordinates shown in figure (5.5b) by propagating the light forward or backwards in time from these transformed coordinates to be synced with the origin's time of zero. So the light's position in the moving frame at the time of zero is

$$\underline{l} = \underline{r} - \underline{c}t \quad (5.12)$$

Where $\underline{r}$ is the light's positions from the Lorentz transformed rest frame coordinates at the Lorentz transformed time t , with $\underline{c}$ being the velocity of light in the moving frame.

What we have learned here can be extended to multiple pulses, which we will do in the next chapter to see what a field of light pulses from a moving source would look like.

5.6 Retarded Propagation of Light in a Source's Moving Frame

There is a delay between when the light is emitted from a source and when it reaches a receiver; this is the propagation time T_p . As a result, there is a difference between where you are seeing moving objects and where they are currently. If the object is moving at velocity $\underline{u}$, you see it at its previous position $-\underline{u}T_p$, where it was positioned at the previous time $-T_p$. We can use the following figure to find the retarded position and time at which the light was emitted, then use this to find the velocity of the light.

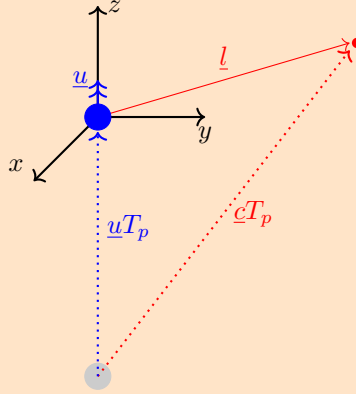


Figure 5.6: **Retarded propagation.** Diagram showing, a source moving at velocity $\underline{u} = (0, 0, u)$, the source is currently at the origin with time $t = 0$, the source had emitted the light from the retarded position $-\underline{u}T_p$ shown in gray (with T_p being the propagation time of the light and source, from the emission), the light was then propagated along $\underline{c}T_p$ shown by red dotted line, to reach position $\underline{l}$ at $t = 0$.

To find the propagated distance of the light, we first have to derive the propagation time. You can skip past this derivation if you would like, as it does not add much to the understanding.

derivation:

The triangle that the three vectors make in the diagram above allows us to get the equation for the velocity of the light as

$$\underline{c}T_p = \underline{u}T_p + \underline{l} \quad (5.13)$$

with $\underline{l} = (l_x, l_y, l_z)$, writing out the vectors and rearranging to get the velocity vector for the light, gives

$$\underline{c} = \frac{1}{T_p} \begin{pmatrix} l_x \\ l_y \\ l_z + uT_p \end{pmatrix} \quad (5.14)$$

The magnitude of the light's displacement is equal to the distance the

light travels in the corresponding propagation time $cT_p = \|\underline{c}T_p\|$. The square of this gives

$$\begin{aligned}(cT_p)^2 &= \|\underline{c}T_p\|^2 \\ &= (l_x^2 + l_y^2 + l_z^2) + u^2 T_p^2 + 2uT_p l_z,\end{aligned}\tag{5.15}$$

rearranging this, we get the quadratic

$$T_p^2 - \left(2\gamma^2 \frac{u}{c^2} l_z\right) T_p - \frac{\gamma^2}{c^2} (l_x^2 + l_y^2 + l_z^2) = 0,\tag{5.16}$$

where $\gamma = 1/\sqrt{1 - u^2/c^2}$. Now taking the positive solution to this, which corresponds to the past time, and making use of the identity

$$\gamma^2 = 1 + \gamma^2 \frac{u^2}{c^2},\tag{5.17}$$

we get the result

$$\begin{aligned}T_p &= \gamma^2 \frac{u}{c^2} l_z + \sqrt{\left(-\gamma^2 \frac{u}{c^2} l_z\right)^2 + \frac{\gamma^2}{c^2} (l_x^2 + l_y^2 + l_z^2)} \\ &= \gamma^2 \frac{u}{c^2} l_z + \frac{\gamma}{c} \sqrt{l_x^2 + l_y^2 + \left(1 + \gamma^2 \frac{u^2}{c^2}\right) l_z^2} \\ &= \gamma^2 \frac{u}{c^2} l_z + \frac{\gamma}{c} \sqrt{l_x^2 + l_y^2 + \gamma^2 l_z^2}\end{aligned}\tag{5.18}$$

This is the time it has taken to propagate the light from where the source was when it was emitted to the current position.

The propagated distance of the light can then be found from the derived propagation time, as

$$\begin{aligned}L_p &= cT_p \\ &= \gamma^2 \frac{u}{c} l_z + \gamma L \\ &= \gamma \left(1 + \frac{u}{c} \frac{\gamma l_z}{L}\right) L\end{aligned}\tag{5.19}$$

With $L = \sqrt{l_x^2 + l_y^2 + \gamma^2 l_z^2}$.

Substituting this into equation (5.14), we can find the light's velocity at $\underline{l}$ to be

$$\underline{c} = \frac{c}{\gamma \left(1 + \frac{u}{c} \frac{\gamma l_z}{L} \right)} \begin{pmatrix} \frac{l_x}{L} \\ \frac{l_y}{L} \\ \gamma \left(\frac{\gamma l_z}{L} + \frac{u}{c} \right) \end{pmatrix} \quad (5.20)$$

If we remove the speed of light from this light's velocity, we obtain its unit vector, with its z-component equal to the cosine of the angle from the z-axis, in this frame with the source moving, giving an equation that will come in useful in the next chapter:

$$\cos \alpha = \frac{\frac{\gamma l_z}{L} + \frac{u}{c}}{1 + \frac{u}{c} \frac{\gamma l_z}{L}} \quad (5.21)$$

Earlier, in the derivation above, you may have noticed that we introduced the γ term. Even though we are not changing frames and we are just looking at the delay in the signal that light has in reaching a particular point. We have made use of the γ term to simplify the equations as we do in special relativity, but we are not invoking special relativity here. I just wanted to emphasize that the γ term plays a role to here too, and it has to be this way to ensure special relativity is self-consistent when switching between frames. We will be looking at this when switching between frames in the next section.

5.7 Light's Propagation when switching from Rest Frame

This section is similar to what we just did, but here we will switch between frames, and that is where the emphasis will be. This section will be used in the next chapter when we have continuous pulses of light in all directions from a source and want to see how this field of light transforms and what it

looks like with a moving source.

Here, we will look at the propagation of light that had been emitted from a source at rest, then switch to a frame where the source is now moving, using the Lorentz transformation of the emitted light's current locations in the rest frame (from $\underline{r}_0$ to $\underline{r}$), but remembering these Lorentz transformed times t are not synchronized with the source's time of zero. So, we will use the moving frame's velocity of light, $\underline{c}$, to propagate back the light from where it is at time t to where it is at the origin's time of zero.

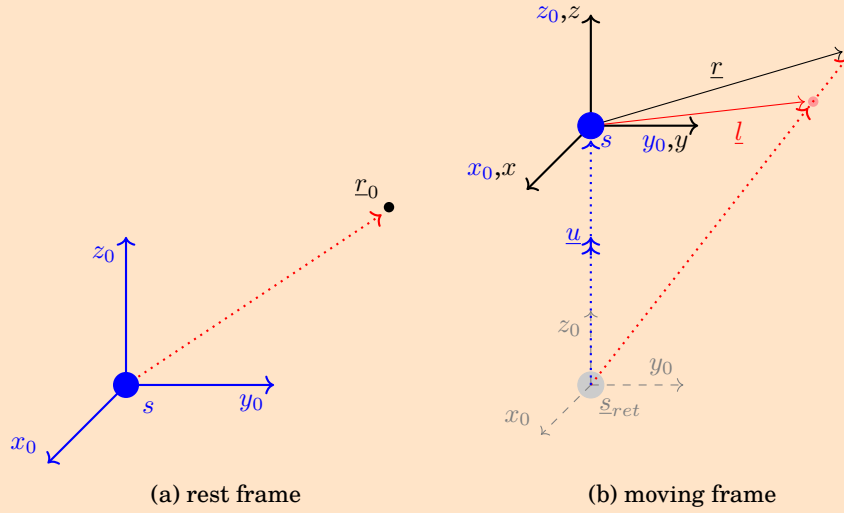


Figure 5.7: Propagation of emitted light in the rest and moving frame. A Diagram showing a source s in blue in its rest frame, Fig (a), with light that has propagated along the red dotted line, to the coordinate $\underline{r}_0$ shown in black, with the current time of $t_0 = 0$. Fig (b) shows the system in a frame with the source and its attached rest axis moving at velocity $\underline{u} = (0, 0, u)$ in the z -direction, with the source currently at the origin in this frame with time $t = 0$. When light was emitted from the source, the source and attached rest axis were at the retarded position $\underline{s}_{ret}$ shown in gray. The emitted light was then propagated along the red dotted line to reach $\underline{l}$ at $t = 0$. It reaches the coordinate $\underline{r}$, which is the Lorentz transform of the rest frame's position at the Lorentz transformed time $t = \gamma \frac{u}{c^2} z_0$.

Now to derive the light's position at time of zero in the moving frame, $\underline{l}$

derivation:

As visualized in figure (5.7) above. If we have a source in its rest frame that had emitted light from the origin and propagated to its current position $\underline{r}_0$ at time $t_0 = 0$. Then using the coordinate transform from equation (2.37) on position $\underline{r}_0 = \underline{l}_0 = (l_{0x}, l_{0y}, l_{0z})$ with time $t_0 = 0$, we get

$$\underline{r} = \begin{pmatrix} l_{0x} \\ l_{0y} \\ \gamma l_{0z} \end{pmatrix}, \quad (5.22)$$

$$\text{at time: } t = -\gamma \frac{v}{c^2} l_{0z} = \gamma \frac{u}{c^2} l_{0z}$$

The light's velocity in the moving frame can be found by using the velocity transform from equation (5.4) and converting the spherical polar coordinate to Cartesian using equation (5.1), giving

$$\underline{c} = \frac{c}{\gamma \left(1 + \frac{u}{c} \frac{l_{0z}}{\|\underline{l}_0\|} \right)} \begin{pmatrix} \frac{l_{0x}}{\|\underline{l}_0\|} \\ \frac{l_{0y}}{\|\underline{l}_0\|} \\ \gamma \left(\frac{l_{0z}}{\|\underline{l}_0\|} + \frac{u}{c} \right) \end{pmatrix} \quad (5.23)$$

Now we have enough to find the position $\underline{l}$ at which the light is when its time is synchronized with the source. We could use the triangle in figure (5.7b), as we did in the previous section, which gives $\underline{l} = \underline{c}T_p - \underline{u}T_p$. But it is easier to use the Lorentz transformed position and time, and then propagate the light back to the time of zero, which is what we did in equation (5.12).

$$\begin{aligned}
\underline{l} &= \underline{r} - \underline{c}t \\
&= \begin{pmatrix} l_{0x} \\ l_{0y} \\ \gamma l_{0z} \end{pmatrix} - \frac{\frac{u}{c} l_{0z}}{1 + \frac{u}{c} \frac{l_{0z}}{\|\underline{l}_0\|}} \begin{pmatrix} \frac{l_{0x}}{\|\underline{l}_0\|} \\ \frac{l_{0y}}{\|\underline{l}_0\|} \\ \gamma \left(\frac{l_{0z}}{\|\underline{l}_0\|} + \frac{u}{c} \right) \end{pmatrix} \\
&= \frac{1}{1 + \frac{u}{c} \frac{l_{0z}}{\|\underline{l}_0\|}} \begin{pmatrix} l_{0x} + \frac{u}{c} \frac{l_{0z}}{\|\underline{l}_0\|} l_{0x} - \frac{u}{c} l_{0z} \frac{l_{0x}}{\|\underline{l}_0\|} \\ l_{0y} + \frac{u}{c} \frac{l_{0z}}{\|\underline{l}_0\|} l_{0y} - \frac{u}{c} l_{0z} \frac{l_{0y}}{\|\underline{l}_0\|} \\ \gamma l_{0z} + \gamma \frac{u}{c} \frac{l_{0z}}{\|\underline{l}_0\|} l_{0z} - \frac{u}{c} l_{0z} \gamma \left(\frac{l_{0z}}{\|\underline{l}_0\|} + \frac{u}{c} \right) \end{pmatrix}
\end{aligned} \tag{5.24}$$

Finally we can cancel out terms and simplify to get

$$\begin{aligned}
\underline{l} &= \frac{1}{1 + \frac{u}{c} \frac{l_{0z}}{\|\underline{l}_0\|}} \begin{pmatrix} l_{0x} \\ l_{0y} \\ \frac{1}{\gamma} l_{0z} \end{pmatrix} \\
&\text{at time } t = 0
\end{aligned} \tag{5.25}$$

This is the position of the light at time zero, the same as the time of the source at the origin. This equation is what we will use to transform the positions of light in the next chapter, when we look at light being continuously emitted from a source, as shown in figure (6.1).

5.8 Summary

In this chapter, we looked at the equations for spherical polar coordinates and, with them, we derived the transform for the velocity of light. We used it to show that the emitted light is angled more towards the direction of a moving source relative than in the source's rest frame, a phenomenon known as relativistic aberration. We also showed that the frequency increases in the direction of motion relative to its rest frame, a phenomenon known as the Doppler effect. Together, these two effects are called relativistic beaming.

After this, we looked at emitted light at time $t_0 = 0$ in the source's rest frame, and how to transform it to a frame where the source is moving, to a time $t = 0$. What we have learned will all come in useful in the next chapter, when we look at a source that continuously emits light in all directions and how this field of light transforms to a frame in which the source is moving.

References

- [2] Albert Einstein et al. "On the electrodynamics of moving bodies". In: *Annalen der physik* 17.10 (1905), pp. 891–921.

Chapter 6

A Point Source: Continuously Emitting Light

This chapter will examine a source that, in its rest frame, continuously emits evenly distributed pulses of light. It gives a "field" of light, as shown in the first diagram in figure (6.1), with the second diagram in the figure showing how this "field" of light would look in a frame where the source is now moving. Equation (5.25) was used to transform each part of the propagating light to where it is at time $t = 0$ (the time that it is synchronized with the source at the origin). The third diagram in the figure shows the rings for each pulse from a moving source. The Doppler effect in each direction is seen in how compact the rings are in each direction, with the aberrational effect on each ring shown by how the light is spread out on it. This combination of the relativistic Doppler and aberration effects together is called relativistic beaming.

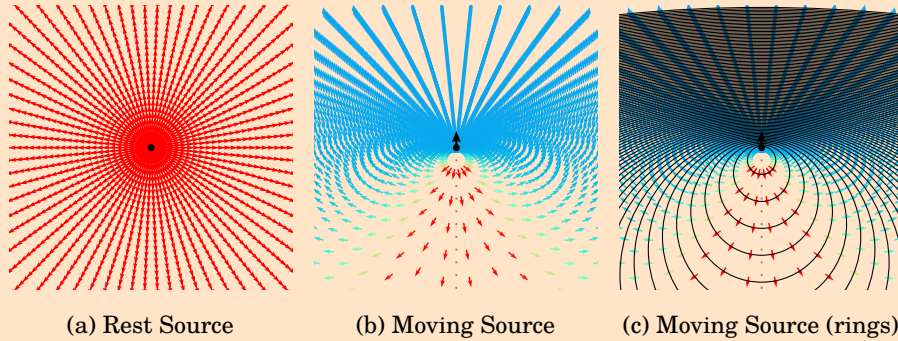


Figure 6.1: **Source's field of Emitted Light.** A diagram showing, in the y-z plane, multiple spherically symmetrical pulses of light from a source at rest (left), and for a source that is moving (middle), the colors show the Doppler effect on the light and the gray dots show the past positions of the source when it had emitted each pulse, there is also a diagram showing the circles of each of the spherical pulses (right), it shows the spherical waves being closer together in the direction of movement.

Now that we have the means to transform this "field" of light, we can examine some of its physical/mathematical properties by considering each property in its rest frame and then determining how it extends to a frame in which it is moving. But first, we will look at how to describe an infinitesimal element on the surface of each spherical pulse. With this infinitesimal element, we will see how the density of the light on each pulse's surface changes when we look at a moving source, along with the Doppler effects on the density of light in the radial direction.

6.1 Surface Element on a Sphere and its Solid Angle

We will again return to spherical polar coordinates, but this time we will look at the differential elements used to describe an infinitesimal surface area on a sphere and, in turn, find out what a solid angle is and how to derive it. We can then find the density of light within this solid angle element. Then, using what we have learned previously about the aberration of light, we can

find how this density of light changes for the case of a moving source. So if we define a sphere with radius r , we can find an infinitesimal portion of the surface of the sphere at angles θ and ϕ as shown in this figure:

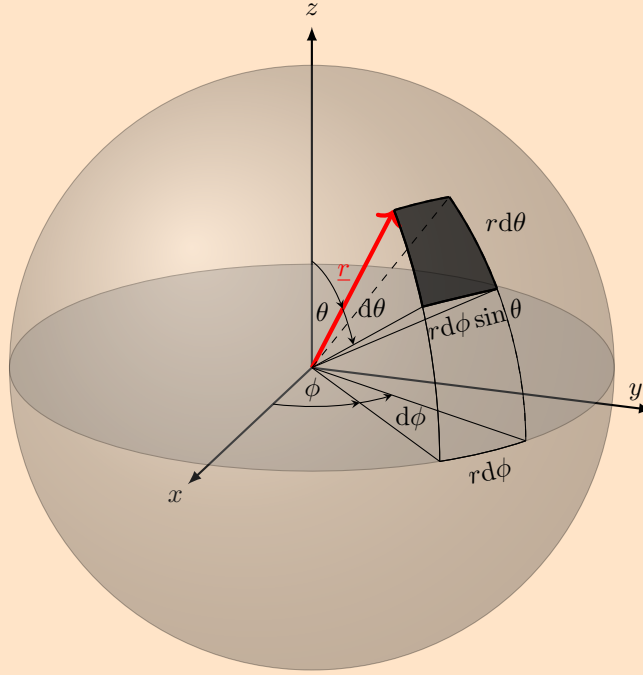


Figure 6.2: Differential surface element on a sphere. This Diagram shows an infinitesimal surface element on a sphere of radius r that spans from θ to $\theta + d\theta$ and from ϕ to $\phi + d\phi$ in spherical polar coordinates. The infinitesimal surface element is equivalent to a rectangle with sides $r d\theta$ and $r \sin \theta d\phi$, giving an infinitesimal surface area of $r^2 \sin \theta d\theta d\phi$.

The surface element of a sphere at a constant radius r , spanning from θ to $\theta + d\theta$ and ϕ to $\phi + d\phi$, is equivalent to a rectangle with sides $r d\theta$ and $r \sin \theta d\phi$ giving the surface area as

$$dA = r^2 \sin \theta d\theta d\phi \quad (6.1)$$

This surface element's size depends on the angle from the z-axis and the distance from the origin, but if we want a quantity that depends only on the angle, we can divide the surface element by r^2 ; this gives what is called

the solid angle. The solid angle element $d\Omega$ is a measure of the amount of the field of view, from the origin, that the surface element covers. For the differential surface element shown in the figure, we have the differential solid angle

$$d\Omega = \frac{dA}{r^2} = \sin \theta d\theta d\phi \quad (6.2)$$

If we were to integrate this over the whole sphere $\int_0^{2\pi} \int_0^\pi \sin \theta d\theta d\phi$, we would get 4π , which is when the surface fully covers the view from the origin. Therefore, the fraction of the view from the origin that is covered by a solid angle Ω is $\Omega/4\pi$.

6.2 Surface Element's Light Density

Here we shall derive the light's density over an infinitesimal surface area perpendicular to the light's propagation, for a moving source. We will do this by first finding the density of light in the solid angle element for a moving source relative to the corresponding solid angle in the rest frame, then multiplying by the dissipation factor to get the density for the surface element, which is the inverse square of the propagated distance of the light from where it was emitted.

derivation:

If we have a source at rest emitting light in all directions, we can say that at certain angles given by the spherical polar coordinates $\theta = \alpha_0$ and $\phi = \varphi_0$, we have a quantity dQ of the emitted light's wavefront that is encompassed in the infinitesimal solid angle

$$d\Omega_0 = \sin \alpha_0 d\alpha_0 d\varphi_0, \quad (6.3)$$

This is the same amount of the light's wavefront that is encompassed by the coinciding solid angle in the moving frame

$$d\Omega = \sin \alpha d\alpha d\varphi. \quad (6.4)$$

We can calculate this element by differentiating both sides of equation (5.5) with respect to α_0 [3], which, after doing so, gives

$$\sin \alpha d\alpha = \frac{1}{\gamma^2 \left(1 + \frac{u}{c} \cos \alpha_0\right)^2} \sin \alpha_0 d\alpha_0 \quad (6.5)$$

now using this and $d\varphi = d\varphi_0$ (as the angle φ_0 is always perpendicular to the motion of the particle and unaffected by the transformation), we have the solid angle in the moving frame given as

$$d\Omega = \frac{1}{\gamma^2 \left(1 + \frac{u}{c} \cos \alpha_0\right)^2} \sin \alpha_0 d\alpha_0 d\varphi_0. \quad (6.6)$$

If there is the same quantity of light dQ in the solid angle element in both frames, then the density of light in the solid angle in each frame is $dQ/d\Omega_0$ and $dQ/d\Omega$ respectively, giving the ratio of the density in the solid angle element in the moving frame relative to the rest frame as

$$D_\sigma = \frac{d\Omega_0}{d\Omega} = \gamma^2 \left(1 + \frac{u}{c} \cos \alpha_0\right)^2. \quad (6.7)$$

By substituting equation (5.21) into equation (5.6), we get $\cos \alpha_0 = \frac{\gamma l_z}{L}$, which allows us to get this in terms of the light's position in its moving frame

$$D_\sigma = \gamma^2 \left(1 + \frac{u}{c} \frac{\gamma l_z}{L}\right)^2. \quad (6.8)$$

with $L = \sqrt{l_x^2 + l_y^2 + \gamma^2 l_z^2}$.

D_σ is the density of light in a solid angle element for a moving source relative to the density of light in the corresponding solid angle element in the rest frame of the source.

At position (l_x, l_y, l_z) , light has propagated a distance L_p from where it was emitted, given in equation (5.19). Its surface area element grows by a factor of L_p^2 , and the light's density in a surface element's area dissipates by the inverse of this:

$$\frac{1}{L_p^2} = \frac{1}{\gamma^2 \left(1 + \frac{u}{c} \frac{\gamma l_z}{L}\right)^2 L^2} \quad (6.9)$$

Multiplying this by the density of light in the solid angle element relative to the rest source, D_σ/L_p^2 , we get the density of light in the surface element given as

$$D_\sigma = \frac{k_\sigma}{L^2}$$

with $L = \sqrt{l_x^2 + l_y^2 + \gamma^2 l_z^2}$

(6.10)

Where k_σ is the proportionality constant.

For a source at rest, $u = 0$, we have $\gamma = 1$, and this equation then gives the inverse square of the distance from the rest source, as

$$D_\sigma = \frac{k_\sigma}{l_x^2 + l_y^2 + l_z^2} \quad (6.11)$$

This inverse square law is what we will get when setting $u = 0$ for all the properties of the light field that we will look at in each section.

$\frac{1}{L^2}$ will appear several times in this chapter for a moving source, and is plotted below for a moving source and a source at rest.

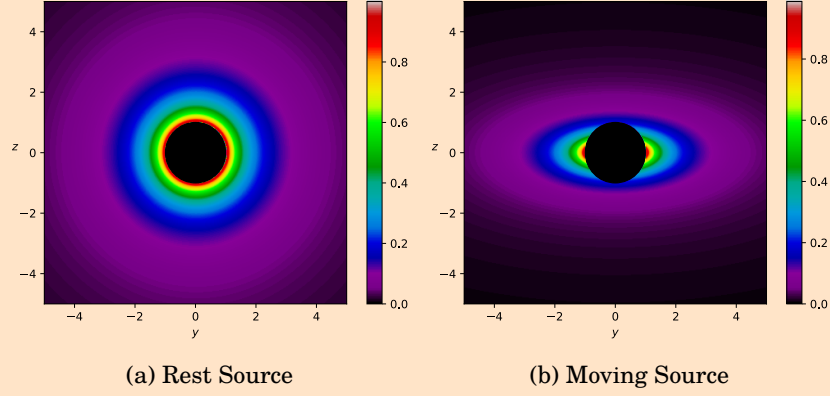


Figure 6.3: **Surface element's light density.** This Figure shows the density of light in a surface element D_σ , at any point in the y-z plane. The value of this goes to infinity as we get closer to the source, so to color the graph nicely, we needed to remove the center, shown by the black circle.

6.3 Density in the direction of propagation of light

For the case where light is being continuously emitted from a source at rest, as shown in figure (6.1a), the light is emitted from the stationary source at a constant rate. Because of this constant rate of emission from the rest source, the light has no change in density in the direction of the light's propagation. The Doppler effect gives the frequency of light from a moving source relative to its rest frame, in which it is constant. The change in density of light in its propagation direction is inversely proportional to this change in frequency, Since this density is constant for a stationary source. Then we have this density for a moving source proportional to the Doppler effect from equation (5.10), which we can get in our current frame's coordinates by substituting in equation (5.21), giving this density for a moving source to be

$$D_c = \gamma \left(1 + \frac{u}{c} \frac{\gamma l_z}{L} \right) k_c \quad (6.12)$$

Where k_c is the constant density of light in the direction of the light's propa-

gation for a stationary source.

6.4 Coordinates used

Previously, we had used a separate notation for the source's rest frame. In this chapter, we will not switch between frames; we will remain in the same frame and change the source's speed u instead. The speed will be set to zero, $u = 0$, to look at the case of a source at rest. We will therefore use the same notation for position coordinates throughout this chapter: (x, y, z) .

We will be substituting these coordinates into equations from previous sections, for the position of light in the moving frame

$$\underline{l} = (l_x, l_y, l_z) = (x, y, z) \quad (6.13)$$

I hope this will not lead to confusion, since we previously used a separate notation for the source's rest frame.

6.5 Volume Density of Light

Here, we will see what the volume density of light is for a moving source. We have the densities of light in the direction of its propagation, D_c , and in the plane perpendicular to it, D_σ , given by equations (6.10) and (6.12).

Giving the volume density to be

$$\begin{aligned} D &= D_\sigma D_c \\ &= \gamma \left(1 + \frac{u}{c} \frac{\gamma z}{L} \right) \frac{k_\sigma k_c}{L^2} \end{aligned} \quad (6.14)$$

This density of light for a moving source at any point is shown in the following diagram: the left figure shows the volume density of light emitted from a source at rest when $u = 0$, demonstrating the inverse square law.

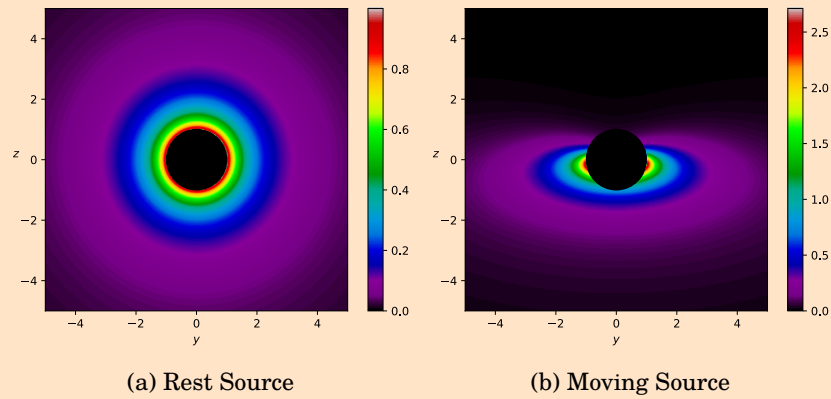


Figure 6.4: **Density of light emitted from a source.** This Figure shows the infinitesimal volume density of the emitted light from a source at any position in the 2D y - z plane, for a moving source and a source at rest. The density at the center of each figure approaches infinity as we get closer to the source, so to color the graph, we needed to remove the center; hence, the black circle.

6.6 Summary

This summary has been left as an exercise for the reader to fill in.

References

- [3] David W Hogg. “Special relativity”. In: *Lecture Notes* (1997), pp. 39–40.

Part IV

Conclusion and Resources

Conclusion

This handbook of special relativity has gone through a lot. However, there is still plenty I have yet to distill, and I will hopefully include more sections when I get the time. For now, I hope this handbook was useful to you. If you are still looking for more in the meantime, there are plenty of external resources in chapter (6.7.2) to expand your knowledge and give you some different perspectives and mental practice.

I will leave you with a final note on gravity and its role in relativity.

6.7 A Note on Gravity

For a reference frame to be inertial there has to be zero acceleration, but the gravity of the earth and other masses accelerates objects towards them. Out in space, we still have this acceleration even though it may be weak, so special relativity can actually only be used as an approximation when modeling a system. Where gravity is stronger or time scales are much longer and these approximations are not good enough, we require an even more complex view of the world and the next level of frame transformations, between two non-inertial frames of reference. This is where general relativity comes in. It is used to explain the redshift of light from stars and also the bending of light around massive stars. It also explains the difference in the flow of time at different strengths of gravity; this gravitational time dilation has to be taken into account by the satellites used by GPS systems. But there are downsides

to general relativity, such as its incompatibility with quantum mechanics, its failure to explain why galaxies rotate much faster than they should based on the visible matter, and its failure to explain why the expansion of the universe is accelerating. It requires inventing new invisible entities to explain these, called dark matter and dark energy (which are vague concepts, as they are whatever new particles or physics are invented that happen to describe some of the discrepancy in the gravity well enough). However, these concepts still fail in many cases. There is also a competitor theory called MOND (Modified Newtonian Dynamics). While it clears up some discrepancies, it fails in many other instances.

So the correct theory and full description of gravity is still to be decided. But if you want to start learning more about gravity, the thought experiment that started general relativity is undeniably a perfect starting point.

6.7.1 Elevator Thought Experiment and the Equivalence Principle

If you were in an elevator in free fall, you would feel weightless, as you would also be free-falling at the same speed, and feel no force from the floor of the elevator into your feet, just like the springs feel no force in the inertial frame in the first diagram of Figure (1.3). You would also feel the same weightlessness if you were in an elevator out in deep space, away from any masses. Even though someone at rest on earth would see you accelerating towards the ground in the free-falling elevator, you would not be able to tell the difference between these two situations from inside the windowless elevators, i.e., the reference frames of both elevators are equivalent. The free-falling elevator feels as though it is at rest or in an inertial frame of reference.

Now, what about an elevator at rest on earth? If a ball is dropped from rest, as shown in figure (6.5), it has an acceleration of g downwards.

Now, what if we apply the same acceleration to the elevator in deep space?

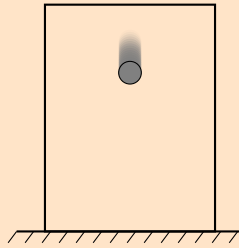
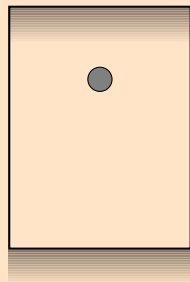
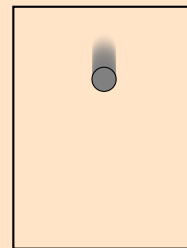


Figure 6.5: An elevator at rest, in earth's non-inertial rest frame, showing a ball accelerating towards the earth due to gravity.

In an inertial frame outside the elevator we would see the elevator accelerate towards the ball, as shown in figure (6.6a). But inside the elevator, in its non-inertial frame, we would see the ball accelerate towards the elevator floor at the same acceleration as the elevator at rest on earth, as shown in figure (6.6a). You would also feel the same weight on your feet. Again, the frames of both of these elevators are equivalent.



(a) Accelerating elevator in space, inertial frame



(b) Accelerating elevator in space, elevator's frame

Figure 6.6: A ball in an accelerating elevator in deep space, with an inertial frame (left) showing a stationary ball with the elevator accelerating upwards, and also the elevators non-inertial rest frame (right), showing the stationary elevator with the ball accelerating downwards.

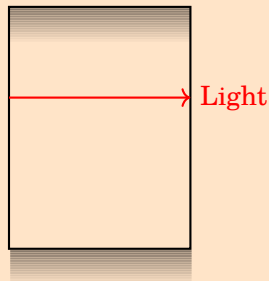
So the mass released from rest behaves identically whether the elevator is accelerating upwards in deep space at g or sitting stationary on the surface

of the earth experiencing a gravitational pull of g .

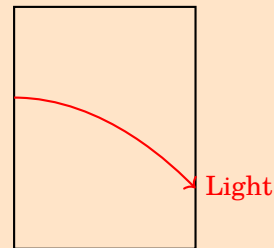
Formally, the gravitational equivalence principle states that there is no local physical experiment you can perform to distinguish between being in a uniform gravitational field and being in a uniformly accelerating reference frame.

Let us now see how this equivalence allows us to deduce that gravity bends the path of light.

If we are in an inertial reference frame outside the elevator in deep space, and shine a light horizontally through a hole in the elevator, it will travel in a straight line across the elevator in this inertial frame. But the elevator is accelerating, and as the light moves across, the elevator rises up more and more quickly, and the light hits the other side of the elevator at a lower point, as shown in figure (6.7a). From the non inertial frame, inside the elevator, we would see the path of light bend downwards and hit the wall at the lower point, as shown in figure (6.7a).



(a) Accelerating elevator in space, inertial frame



(b) Accelerating elevator in space, elevator's rest frame

Figure 6.7: An accelerating elevator in deep space, with light traveling across it, with an inertial frame (left) showing a straight light path across the elevator while the elevator accelerates upwards, and also the elevator's non-inertial rest frame (right), showing the stationary elevator with the light's path bent downward.

The light's path is bent in the non inertial frame, but from the equivalence principle, the frame inside an elevator at rest on earth is equivalent, since it

experiences the same acceleration due to gravity. So we would then expect to find that gravity does, in fact, bend the light by the same amount, shown in figure (6.8). This has been experimentally verified: light does indeed bend around heavy masses such as stars, black holes, and to a lesser extent, planets.

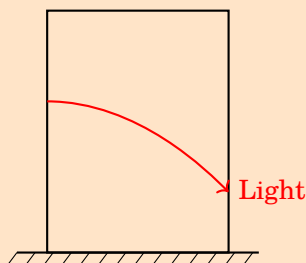


Figure 6.8: An Elevator at rest on earth, in the earth's non-inertial frame, with light being bent towards the ground, as it is equivalent to the scenario with the accelerating elevator in deep space.

In the real world, the gravitational field in the elevator on earth is not perfectly uniform, but it is a good approximation. The top of the elevator is further from earth and so gravity is a slightly weaker than at the bottom. Also, for a spherical mass, the gravitational field spreads out spherically, creating a negligible difference in the gravitational strength in the center of the elevator compared to the walls; this is called a tidal force. These are negligible effects for our thought experiment, but might be significant when modeling other systems.

In general relativity, the local speed of light is always equal to c according to the observer who is there. But when viewed from a different gravitational strength, it appears to be moving at a different speed. This effect is tied to gravitational time dilation, which we will give a note about next.

6.7.2 Gravitational Time Dilation

We have seen the time dilation equation in special relativity between two reference frames, but gravity also has a time dilation associated with its strength. The stronger the gravity at a point, the slower time passes relative to an inertial reference frame that doesn't experience any gravitational force.

$$\Delta t_g = \Delta t_\infty \sqrt{1 - \frac{v_{esc}^2}{c^2}} \quad (6.15)$$

Where v_{esc} is the minimum velocity an object has to be traveling away from a mass to escape its orbit, and Δt_g is the change in time within a gravitational well compared to the change in time at an infinite distance away Δt_∞ . This is the same as the time dilation in special relativity as seen in equation (2.15) if we were to use $v = v_{esc}$. The escape velocity is also the same velocity (but in the opposite direction) that a particle would have if it started at rest an infinite distance away and reached the current position in a stationary mass's gravitational field. There is an interpretation of general relativity called the "river model" [6], which suggests that gravity and its effects are due to space itself falling inwards at this velocity at each point. It also extended this to rotating masses. This model is an interpretation that can be useful for learning general relativity, but it is a niche interpretation. So, you can keep an open mind about it and remain skeptical about whether the interpretation is worthwhile.

This concludes The Handbook of Special Relativity, but please check out the other resources on this topic that I have provided in the next bit.

Resources

Here are some useful resources for learning different aspects of special relativity, including spacetime diagrams which I choose not to use as a learning aid in this handbook.

There is one thing that you should take into account when using these resources compared to this handbook, and that is that in this handbook we have chosen for the primed frame to be moving in the z-direction throughout, as it was needed when in came to spherical coordinates, whereas it is more common to have the primed frame going in the x-direction. So you just need to swap the x and z coordinates in transformation equations to get the same transformations.

Books:

- *Relativity Visualized* by Lewis Carroll Epstein

Sets of video lessons:

- *Intro to special relativity* by minutephysics
- *Special relativity* by minutephysics
- *Special relativity, an introduction without calculus* by James Battat
- *Relativity* by eigenchris

Single videos:

-
- *Visualizing Time Dilation by ScienceClic*
 - *Special Relativity by ScienceClic*
 - *Spacetime rotations, understanding Lorentz transformations by ScienceClic*

Webpage of interactive diagrams:

- https://javalab.org/en/category/measure_en/relativity_en/
- <https://brilliant.org/courses/special-relativity/>

Spacetime Diagrams:

- <https://ibises.org.uk/Minkowski.html>
- https://tikz.net/relativity_minkowski_diagram/
- <https://prideout.net/blog/relativity/>

Definitions

Mathematical Framework

Reference frame: An abstract coordinate system, consisting of three spatial and one time axis. The origin, orientation, and scale of its axis are specified by a set of points in space. The purpose of it is to provide a standardized means of defining the position and orientation of objects at any instant in time.

Inertial reference frame: A reference frame that is not undergoing any acceleration. An observer's reference frame is inertial if they do not feel a force acting on them.

Proper/Rest frame: This is the reference frame of the observer or object itself.

Primed frame: A reference frame that is moving relative to the current frame.

Initial frame: The first reference frame that we are comparing the current or primed reference frame to.

Moving frame: An inertial reference frame that is moving relative to the rest frame.

frame velocity: The velocity of a reference frame relative to the current reference frame.

Frame transform: The changing of coordinates and other quantities in a reference frame into their corresponding values in another reference frame.

Galilean/Classical frame transform: Transformation of coordinates and physical quantities according to Newtonian physics between two frames of reference.

Relativistic frame transform: The changing of the spatial coordinates, the time coordinate, and other physical quantities into their corresponding values in a different inertial reference frame using equations from special relativity.

Observer: An entity/thing with an associated frame of reference/system of coordinates, with respect to which you can measure an object's position, orientation, shape, signal emission and other properties.

Event: An occurrence that can be specified by a unique combination of the three spatial and one time coordinates. These four coordinates, together, form a point in spacetime.

Simultaneity: Within a reference frame, two or more events are simultaneous if they occur at the same time within it. Classically, if any two events are simultaneous in one reference frame, they are also simultaneous in another. This is not true in special relativity, and the order of events can depend on the frame of reference.

Retarded position: The previous position of a source, that an observer/reciever currently sees it at, due to the time delay between when a moving source emitting a light signal and when the observer/reciever detects it.

Retarded time: The previous time when a source was at its retarded position (the position it emitted the light signal that is currently being observed by the observer).

Retarded coordinates: The retarded positions of an initial reference frame's coordinates within a

second reference frame, at their retarded times.

The Mechanics

Aether: A proposed medium that filled the vacuum of space that light propagated through.

Vacuum: Space that contains no matter.

Lorentz length contraction: The shortening of a moving object along the direction of its motion relative to an observer (compared to the length of object at rest).

Time dilation: The slowing of the passing of time for objects in motion relative to an observer, compared to the passing of time for the observer.

Doppler effect: The change of a wave's frequency due to the motion of its source relative to an observer. This is due to the bunching up of the wave in the direction of movement of the source and the stretching of the wave in the opposite direction, as well as the dilation of time between the emission of the crests of the emitted wave.

Aberration of light: The changing of the direction of the propagation of light when changing to another reference frame.

Relativistic beaming: The accumulation of the Doppler effect and aberration of light, and how it affects the amount of light and its frequency emitted at different angles from a moving source. This effect can be extended to any wave or field emitted at the speed of light from a moving source.

Retarded field: A moving source's field, where each point in the field has had its field propagated at the speed of light from the source's retarded position (i.e. the previous position of the source where

it had emitted that part of the field from).

Field: A physical quantity that has a value (or a set of values) assigned to every point in space and time.

Light year: The distance light travels in one year (in a vacuum).

Definitions for Later Chapters

Lorentz Invariant/Frame independent: A quantity or law of physics is Lorentz Invariant/Frame independent, if it remains unchanged when transforming coordinates and other quantities to any other inertial reference frame.

Postulates: A statement that is assumed to be true and is used as a basis for reasoning for a theory.

Flux: A measure of how much of a quantity is passing through a given surface in a given time (flowrate through a surface)

Perceived/Apparent Velocity: The velocity an object appears to have due to the time delay in the signal of light from it, as the time delay from each position along a source's path is different due to having different distances the signal needs to travel to observer (It is the consequence of the time difference in the propagation time of the light signal from the different positions of a source's path).

Rest mass: The mass of a particle in its own rest frame.

Rest mass energy: The energy a particle has due to its rest mass (given by its rest mass times speed of light squared).

Math Terms Dictionary

(γ) : the gamma factor, a relativistic factor used to simplify equations, it is given by $1/\sqrt{1 - v^2/c^2}$ which determines how much of an effect the transform of the coordinates deviates from the classical transform. It is only significant at relative frame speeds, v , close to that of light, c .

(x) : the x-coordinate

(z) : the z-coordinate

(y) : the y-coordinate

(x') : the corresponding x-coordinate in the primed frame

(z') : the corresponding z-coordinate in the primed frame

(y') : the corresponding y-coordinate in the primed frame

(v) : The speed of the second reference frame, primed frame, relative to the first reference frame.

$(\underline{r})$: $\underline{r} = (x, y, z)$, the position vector coordinates of an event in the current inertial reference frame.

$(\vec{o}o')$: the position of the primed axis' origin relative to the proper axis' origin, (given in the proper frame)

$(\vec{o'}e)$: the position of the event relative to the primed frame's origin (in proper frame)

(o) : the position of the proper axis' origin, given by the set of spatial and time coordinates.

(o') : the position of the primed axis' origin, given by the set of spatial and time coordinates.

(t) : the time coordinate in current frame

$(\underline{e})$: $\underline{e} = (x, y, z, t)$, an event in the proper frame, given by the set of spatial and time coordinates.

$(\underline{e}')$: $\underline{e}' = (x', y', z', t')$, an event in the primed frame, given by the set of spatial and time coordinates.

(t') : the time coordinate in primed frame

$(\underline{r}')$: $\underline{r}' = (x', y', z')$, the position vector coordinates of an event in another

inertial reference frame, referred to as the primed frame, and denoted with an apostrophe '. This position vector corresponds to the position vector of the event in the initial reference frame.

(c) : The speed that light travels in empty space when it is not impeded by matter and fields.

(Δ) : The interval in the corresponding variable

(d_m) : the distance between mirrors

(u) : the speed of object in current frame, which in this book we will sometimes specifically choose to be in the z-direction

(z_1) : the z-coordinate of first event, in initial frame

(z_2) : the z-coordinate of second event, in initial frame

(Z) : the distance between light bulb and front mirror, in rest frame

(z'_1) : the z-coordinate of first event, in primed frame

(z'_2) : the z-coordinate of first event, in primed frame

(Z') : the distance between light bulb and front mirror, in moving frame

(t'_1) : the time coordinate of first event, in primed frame

(t'_2) : the time coordinate of second event, in primed frame

(x_1) : the x-coordinate of first event, in initial frame

(y_1) : the y-coordinate of first event, in initial frame

(x_2) : the x-coordinate of second event, in initial frame

(y_2) : the y-coordinate of second event, in initial frame

(x'_2) : the x-coordinate of second event, in primed frame

(x'_1) : the x-coordinate of first event, in primed frame

(y'_2) : the y-coordinate of second event, in primed frame

(y'_1) : the y-coordinate of first event, in primed frame

(t_2) : the time coordinate of second event, in initial frame

(t_1) : the time coordinate of first event, in initial frame

($\frac{dr}{dt}$) : the infinitesimal change in position over infinitesimal change in time of a particle

(d) : the infinitesimal change in the corresponding variable

(u'_x) : the speed of object in z-direction, in primed frame

(u'_y) : the speed of object in z-direction, in primed frame
 (u'_z) : the speed of object in z-direction, in primed frame
 $(\underline{u}')$: $\underline{u}' = (u'_x, u'_y, u'_z)$, the velocity vector of object in primed frame
 (u_z) : the speed of object in z-direction, in current frame
 (u_x) : the speed of object in x-direction, in current frame
 (u_y) : the speed of object in y-direction, in current frame
 $(\underline{u})$: $\underline{u} = (u_x, u_y, u_z)$, the velocity vector of object in current frame
 $(\underline{v})$: The velocity of the second reference frame, primed frame, relative to the first reference frame, with v being the speed and $\hat{v}$ the directional unit vector, $\mathbf{v} = v \cdot \hat{v}$.
 $(\underline{a}')$: the acceleration vector in primed frame
 (a_x) : the acceleration in x direction, in primed frame
 (a_z) : the acceleration in z direction, in primed frame
 (a_y) : the acceleration in y direction, in primed frame
 $(\underline{a})$: $\underline{a} = (a_x, a_y, a_z)$, the acceleration vector in initial frame
 $(\hat{v})$: the unit vector of the velocity of the frame relative to the proper/initial frame
 (u_s) : the speed of the signal, in the initial frame
 (t_r) : the time the signal is received, in the initial frame
 (t'_r) : the position of particle b were the signal is received, in the primed frame
 (z_r) : the position of particle b were the signal is received, in the initial frame
 (c') : the speed of light in primed frame, equal to c , the speed of light in any frame
 $(\underline{c}')$: the velocity of light in primed frame
 (S) : the space-time coordinate, in initial frame
 (S') : the space-time coordinate in primed frame
 $(\square)$: the d'Alembert operator in initial frame
 $(\square')$: the d'Alembert operator in primed frame
 (q) : This represents a particle q
 (S_q) : The spacetime coordinate of particle q

(t_q) : The time coordinate of particle q
 (x_q) : The x-coordinate of particle q
 (y_q) : The y-coordinate of particle q
 (z_q) : The z-coordinate of particle q
 (γ_u) : The gamma factor γ between the current frame and particles rest frame, given by $1/\sqrt{1 - u_q^2/c^2}$. It is only significant at relative particle speeds, u_q , close to that of light, c .
 (u_q) : the speed of particle q
 (u_{qx}) : the speed of particle q in the x-direction
 (u_{qy}) : the speed of particle q in the y-direction
 (u_{qz}) : the speed of particle q in the z-direction
 (m) : the mass of a particle
 (E_0) : The energy a particle has at rest, corresponding to the mass has
 (p) : the momentum of a particle
 (p_{after}) : the total momentum in an isolated system after a certain amount of time
 (p_{before}) : the initial total momentum in an isolated system
 (E_{after}) : the total energy in an isolated system after a certain amount of time
 (E_{before}) : the initial total energy in an isolated system
 $(\underline{p})$: $\underline{p} = (p_x, p_y, p_z)$, the momentum vector, in the initial frame
 $(\gamma_{u'})$: The gamma factor γ between the primed frame and the particles rest frame, given by $1/\sqrt{1 - u_q'^2/c^2}$. It is only significant at relative particle speeds, u_q' , close to that of light, c .
 (u') : the speed of the object in the primed frame
 (p'_y) : the momentum in the y-direction, in the primed frame
 (p_y) : the momentum in the y-direction, in the initial frame
 (p'_z) : the momentum in the z-direction, in the primed frame
 (p_z) : the momentum in the z-direction, in the initial frame
 (p_{zafter}) : the total momentum in an isolated system in the z-direction after a certain amount of time

$(p_{zbefore})$: the initial total momentum in an isolated system in the z-direction before
 (E_{ph}) : the total energy of a photon
 (γ_c) : the gamma factor with $v = c$ making it an infinity large number
 (m_{ph}) : the rest mass of a photon, it is shown that the mass must be zero
 (m_{eff}) : a hypothetical effective mass a photon has, given by p_{ph}/c
 (p_{ph}) : the momentum of a photon
 (E_p) : the energy corresponding to the momentum of a particle
 (θ_{ret}) : The angle from the origin between the z-axis and the retarded position, in the initial frame
 $(\underline{r}_{ret})$: the retarded position of an object in the initial frame, it is the position the object was when it emitted the light signal that the observer at the origin is now receiving
 (α) : the angle between the z-direction and the direction the light is propagating, in the initial frame
 $(\underline{r}'_{ret})$: the retarded position of an object in the primed frame, it is the position the object was when it emitted the light signal that the observer at the origin is now receiving
 $(\underline{r}'_{t'=0})$: the position of an object in the primed frame at the current time of the origin $t' = 0$
 (θ'_{ret}) : the angle between the z-axis and the retarded position, in the primed frame
 (t_{ret}) : the retarded time, the time when the object was at its retarded position, in the initial frame
 $(\underline{c})$: the velocity vector of light, it has a magnitude of c
 $(\hat{c})$: the unit directional velocity vector of light, it has a magnitude of 1
 (t'_{ret}) : the retarded time, the time the object was at its retarded position, in the primed frame
 (π) : a constant equal to 3.14159...
 $(\hat{r}_{ret})$: the unit directional vector for $\underline{r}$
 (N) : a negligibly small term

$(\underline{u}_{app})$: the apparent velocity particle due to the delay in the light signal from different points in its path
 $(\underline{c}_{app})$: the apparent velocity of light depending on its direction relative to observer, due to the delay in the signal from different points in its path
 $(\underline{u}_{ret})$: the velocity of the particle at the retarded time
 (r) : $r = \|\underline{x}\|$, the spherical polar coordinate corresponding to the magnitude of the vector
 (ϕ) : the spherical polar coordinate corresponding to the angle from the x-axis to the vector, in the x-y plane
 (θ) : the spherical polar coordinate corresponding to the angle from the z-axis to the vector
 (α_0) : the angle between the z-direction and the direction the light is propagating, in the rest frame of the source
 (φ_0) : the angle in the x-y plane between the x-axis and the direction the light is propagating, in the rest frame of the source
 $(\underline{c}_0)$: the velocity vector of the emitted light in the rest frame of its source, it has a magnitude of c
 $(\hat{\underline{c}}_0)$: the directional velocity vector of the emitted light in the rest frame of its source, it has a magnitude of 1
 (φ) : the angle in the x-y plane between the x-axis and the direction the light is propagating, in the current frame with source moving
 (t_0) : the time in the rest frame of a particle or source
 (λ_0) : length the light wave has traveled from the source in the time t_0 , in the source's rest frame
 (λ) : the distance of the light wave from the source's current position at time t_0 , in the current frame with the source moving
 (ν) : the frequency of light, in the current frame with the source moving
 (ν_0) : the frequency of light, in the source's rest frame
 (ν') : the frequency of light, in a primed frame
 (α') : the angle between the z-direction and the direction the light is propagating, in the primed frame

$(\underline{l}) : (l_x, l_y, l_z)$, the light's position vector, in the current frame with source moving
 $(T_p) :$ the time the object and the light took to propagate from this retarded position to where they currently are, equal to the negative of the retarded time $T_p = -T_{ret}$
 $(L) :$ a factor used to simplify equations $L = \sqrt{l_x^2 + l_y^2 + \gamma^2 l_z^2}$
 $(l_x) :$ the light's x-coordinate, in the current frame with source moving
 $(l_y) :$ the light's y-coordinate, in the current frame with source moving
 $(l_z) :$ the light's z-coordinate, in the current frame with source moving
 $(L_p) :$ the distance light has propagated in the propagation time $T_p = -T_{ret}$
 $(r_0) :$ $r_0 = (x_0, y_0, z_0)$ a position in the rest frame of the source where the light is at time $t_0 = 0$, giving $r_0 = \underline{l}_0$
 $(y_0) :$ a position's y-coordinate in the source's rest frame, where the light is at time $t_0 = 0$, giving $r_0 = \underline{l}_0$
 $(z_0) :$ a position's z-coordinate in the source's rest frame, where the light is at time $t_0 = 0$, giving $r_0 = \underline{l}_0$
 $(x_0) :$ a position's x-coordinate in the source's rest frame, where the light is at time $t_0 = 0$, giving $r_0 = \underline{l}_0$
 $(s) :$ the letter represents a source
 $(\underline{s}_{ret}) :$ the source's retarded position in the current frame with the source moving, it is the position the source was when it emitted the light
 $(\underline{l}_0) :$ $\underline{l}_0 = (l_{0x}, l_{0y}, l_{0z})$, the position of light at time $t_0 = 0$ in the rest frame of the source
 $(l_{0x}) :$ the x-coordinate of light at time $t_0 = 0$ in the rest frame of the source
 $(l_{0y}) :$ the y-coordinate of light at time $t_0 = 0$ in the rest frame of the source
 $(l_{0z}) :$ the z-coordinate of light at time $t_0 = 0$ in the rest frame of the source
 $(\Omega) :$ the solid angle in the current frame, it is a measure of the amount of the field of view
 $(Q) :$ a quantity of the emitted light's wavefront that is encompassed in the infinitesimal solid angle, in the rest frame of the source

(Ω_0) : the solid angle in the sources rest frame, it is a measure of the amount of the field of view

(D_σ) : the ratio of the density in the solid angle element in the moving frame relative to the rest frame

(k_σ) : a proportionality constant

(D_c) : the density of light in its direction of propagation, in the current frame

(k_c) : a proportionality constant

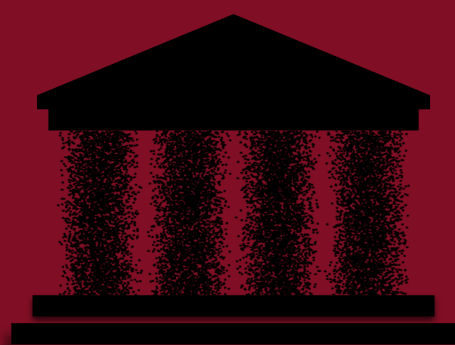
(D) : the volume density of light, in the current frame

(A) : surface area on the sphere corresponding to the given solid angle

Bibliography

- [1] Morley Williams Michelson Abraham. “On the Relative Motion of the Earth and the Luminiferous Ether”. In: *American Journal of Science* 34 (1887).
- [2] Albert Einstein et al. “On the electrodynamics of moving bodies”. In: *Annalen der physik* 17.10 (1905), pp. 891–921.
- [4] Lewis Carroll Epstein. *Relativity Visualized*. Insight Press, 1985. ISBN: 9780935218039.
- [5] Richard P. Feynman, Robert B. Leighton, and Matthew Sands. “The Feynman Lectures on Physics”. In: vol. 1. Section title: The Momentum of Light. Addison-Wesley, 1963. Chap. 34, Section 34–9. URL: https://www.feynmanlectures.caltech.edu/I_34.html#Ch34-S9.
- [6] Andrew J. S. Hamilton and Jason P. Lisle. “The river model of black holes”. In: *American Journal of Physics* 76.6 (June 2008), pp. 519–532. ISSN: 1943-2909. DOI: 10.1119/1.2830526. URL: <http://dx.doi.org/10.1119/1.2830526>.
- [3] David W Hogg. “Special relativity”. In: *Lecture Notes* (1997), pp. 39–40.
- [7] Tom Roberts and S. Schleif. *What is the experimental basis of Special Relativity?* UC Riverside Department of Mathematics. Nov. 3, 2017. URL: <https://math.ucr.edu/home/baez/physics/Relativity/SR/experiments.html> (visited on 11/13/2025).

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- [8] Wikipedia contributors. *Tests of relativistic energy and momentum*. Wikipedia, The Free Encyclopedia. Sept. 28, 2025. URL: https://en.wikipedia.org/w/index.php?title=Tests_of_relativistic_energy_and_momentum&oldid=1313924927 (visited on 11/13/2025).



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